

B.Sc.
COMPUTER SCIENCE

— FIRST YEAR —

COMPUTER NETWORKING

— AND INTERNET —

— Complete Notes for All Units —



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EXAM ORIENTED NOTES



DIAGRAMS & TABLES



CONCEPTS SIMPLIFIED



Subject :
Computer Networking



Notes By :
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Module No. 1

Basics of Computer Network

1.1 Computer Networking

Introduction

The rapid growth of Information and Communication Technology (ICT) has changed the way people communicate, work, learn, and share information. In today's digital world, computers are no longer used as standalone machines. Instead, they are connected with other computers and devices to exchange information and share resources efficiently. This interconnection of computers is known as **Computer Networking**.

Computer networking has become an essential part of our daily lives. Whether we are browsing the Internet, sending emails, attending online classes, using social media, transferring money through online banking, shopping online, or streaming videos, all these activities are possible because of computer networks. Today, almost every organization, including schools, colleges, banks, hospitals, industries, government offices, and multinational companies, depends on computer networks for its day-to-day operations.

Computer Networking

Computer Networking is the process of connecting two or more computers and other digital devices through wired or wireless communication links so that they can exchange data, share resources, and communicate with each other by following standard communication protocols.

A computer network enables connected devices to work together as a single system. Instead of working independently, the devices can share information, access common resources, and communicate quickly and efficiently.

Objectives of Computer Networking

The primary objectives of computer networking are:

- To enable communication between computers and users.
- To share hardware resources such as printers, scanners, and storage devices.
- To share software applications and data among multiple users.
- To provide fast and reliable information exchange.
- To improve collaboration among users working at different locations.
- To reduce operational costs through resource sharing.
- To provide centralized management of data and network resources.

Basic Working of a Computer Network

A computer network works by connecting multiple computers and devices through communication media such as cables or wireless technology. When a user sends data from one computer, the information is transmitted through the communication medium to the destination device. During this process, networking devices help direct the data to the correct destination, while communication protocols ensure that the data is transmitted accurately and securely. The receiving device accepts the transmitted data and makes it available to the user.

This entire process takes place within a very short time, allowing users to communicate and share information efficiently.

Basic Components of a Computer Network

Every computer network consists of several basic components that work together to establish communication.

Computers (Nodes): These are the devices that send, receive, and process data. Desktop computers, laptops, servers, smartphones, and tablets are common examples of network nodes.

Communication Medium: It is the path through which data travels from one device to another. The medium may be wired or wireless.

Networking Devices: Devices such as switches, routers, bridges, repeaters, and gateways help establish communication between different devices in a network.

Communication Protocols: Protocols are a set of rules that control the transmission and reception of data between devices connected to a network.

Note: The communication media, networking devices, and communication protocols will be studied in detail in the subsequent chapters.

Importance of Computer Networking

Computer networking has become an essential technology because it enables efficient communication and resource sharing. It allows users to access information quickly, collaborate with others, and perform various online activities without geographical limitations. Networking also improves productivity, simplifies data management, and supports modern digital services used in education, healthcare, banking, business, and government sectors.

Real-Life Example

Consider a college computer laboratory where all computers are connected through a network. Students can access the Internet, use a common printer, share files with teachers, and store data on a central server. Instead of each computer working independently, all systems work together through the network, making resource sharing and communication faster and more efficient.

Simple Computer Network Diagram

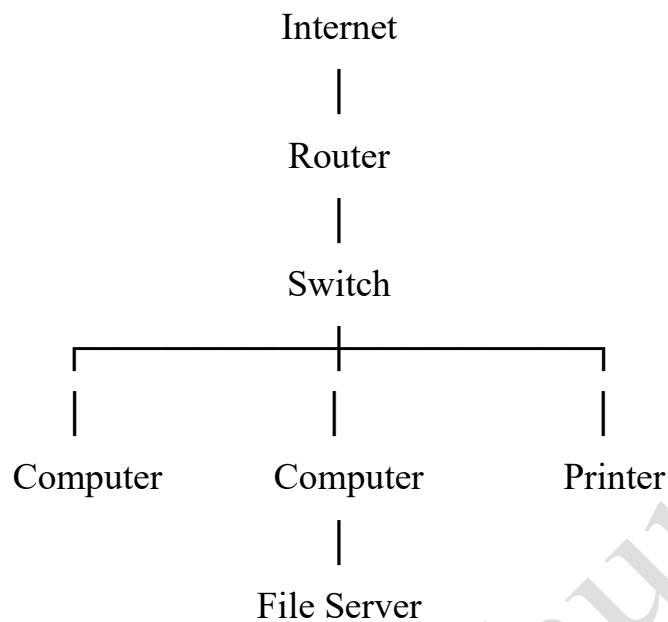


Figure 1.1: A simple computer network connecting multiple devices for communication and resource sharing.

1.2 Signals – Analog and Digital Signals

Introduction

In a computer network, data cannot be transmitted directly from one device to another. Before transmission, the data must be converted into a form that can travel through a communication medium such as a cable or a wireless channel. This form of data is called a **signal**. A signal carries information from the sender to the receiver and plays a vital role in data communication.

Whenever we send an email, browse a website, make a video call, or transfer a file over a network, the information is transmitted in the form of signals. Depending on how the information is represented, signals are classified into two types: **Analog Signals** and **Digital Signals**.

Signal

A **signal** is an electrical or electromagnetic representation of data that carries information from one device to another through a communication medium.

In data communication, a signal acts as a carrier that transports data from the sender to the receiver.

Types of Signals

Signals are mainly classified into two types:

1. Analog Signal
2. Digital Signal

Analog Signal

An **Analog Signal** is a continuous signal that changes smoothly with time. It can have an infinite number of values within a given range. Analog signals are generally represented by a **sine wave** because their amplitude varies continuously.

Analog signals are commonly used to transmit natural information such as human voice, sound, temperature, pressure, and other physical quantities. Since these signals are continuous in nature, they are more susceptible to noise and interference during transmission, especially over long distances.

Characteristics of Analog Signal

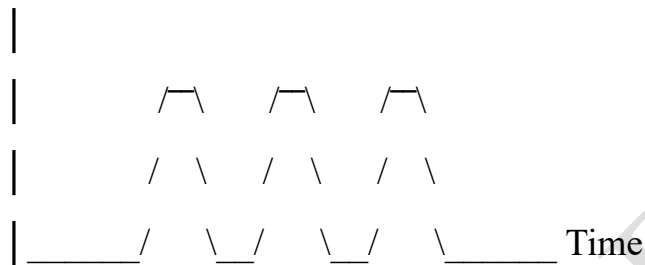
- Continuous in nature.
- Has an infinite number of possible values.
- Represented by a sine wave.
- Smooth variation in amplitude and frequency.
- More affected by noise and distortion.
- Signal quality decreases over long distances.

Applications of Analog Signal

- Traditional telephone systems
- AM and FM radio broadcasting
- Traditional television broadcasting
- Audio recording systems
- Medical monitoring instruments

Analog Signal Diagram

Amplitude



Digital Signal

A **Digital Signal** is a discrete signal that represents information using two fixed levels, **0 (Low)** and **1 (High)**. These two values are called **binary digits (bits)**.

Unlike analog signals, digital signals do not vary continuously. They change between fixed voltage levels, making them more reliable for data communication. Digital signals are less affected by noise and can be regenerated accurately without losing the original information. Therefore, they are widely used in modern computer networks and digital communication systems.

Characteristics of Digital Signal

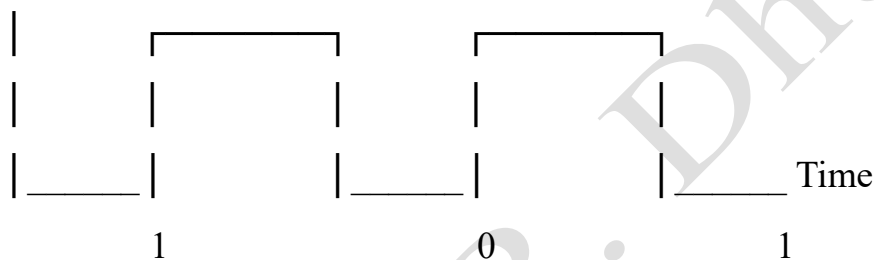
- Discrete in nature.
- Uses only two binary values: 0 and 1.
- Represented by square waves.
- Less affected by noise.
- Easy to regenerate accurately.
- Suitable for high-speed and long-distance communication.

Applications of Digital Signal

- Computer networks
- Internet communication
- Mobile communication
- Digital television
- Optical fiber communication
- Cloud computing
- Digital storage devices

Digital Signal Diagram

Amplitude



Difference Between Analog Signal and Digital Signal

Analog Signal	Digital Signal
Continuous signal.	Discrete signal.
Has an infinite number of values.	Has only two values (0 and 1).
Represented by a sine wave.	Represented by a square wave.
More affected by noise and interference.	Less affected by noise and interference.
Difficult to regenerate accurately.	Easy to regenerate accurately.
Signal quality gradually decreases over long distances.	Signal can be regenerated, maintaining its quality over long distances.

Mainly used in traditional communication systems.	Mainly used in modern computer networks and digital communication systems.
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Why Digital Signals are Preferred in Computer Networks

Digital signals are preferred in modern computer networks because they provide reliable and accurate communication. They are less affected by noise, support high-speed data transmission, and can be regenerated without significant loss of information. Digital signals also make error detection and error correction easier, making them ideal for Internet communication, cloud computing, and other modern networking technologies.

Real-Life Example

When a person speaks during a mobile phone call, the voice is produced as an analog signal. The smartphone converts this analog voice into a digital signal before transmitting it through the mobile network. At the receiver's end, the digital signal is converted back into an analog signal so that the other person can hear the voice clearly. This process enables efficient and reliable communication in modern digital networks.

1.3 Parallel and Serial Transmission Mode

Introduction

In a computer network, data is transmitted from one device to another through a communication medium. The method used to transmit data is known as the **transmission mode**. The transmission mode determines how data bits are sent from the sender to the receiver. An appropriate transmission mode helps in achieving efficient, reliable, and accurate data communication.

Based on the way data bits are transmitted, transmission modes are classified into two types:

1. Parallel Transmission
2. Serial Transmission

Parallel Transmission

Parallel Transmission is a transmission mode in which multiple bits of data are transmitted simultaneously using multiple communication lines. Each bit is transmitted through a separate communication line at the same time.

For example, if one byte consists of 8 bits, all eight bits are transmitted simultaneously using eight separate lines. Since several bits are transmitted together, parallel transmission provides high-speed data transfer. However, it is suitable only for short-distance communication because synchronization between multiple lines becomes difficult over longer distances.

Characteristics of Parallel Transmission

- Multiple bits are transmitted simultaneously.
- Uses multiple communication lines.
- Provides high-speed data transmission.
- Suitable for short-distance communication.
- Requires more cables and connectors.

Advantages of Parallel Transmission

- Faster data transmission over short distances.
- Suitable for transferring large amounts of data.
- Simple communication between nearby devices.

Disadvantages of Parallel Transmission

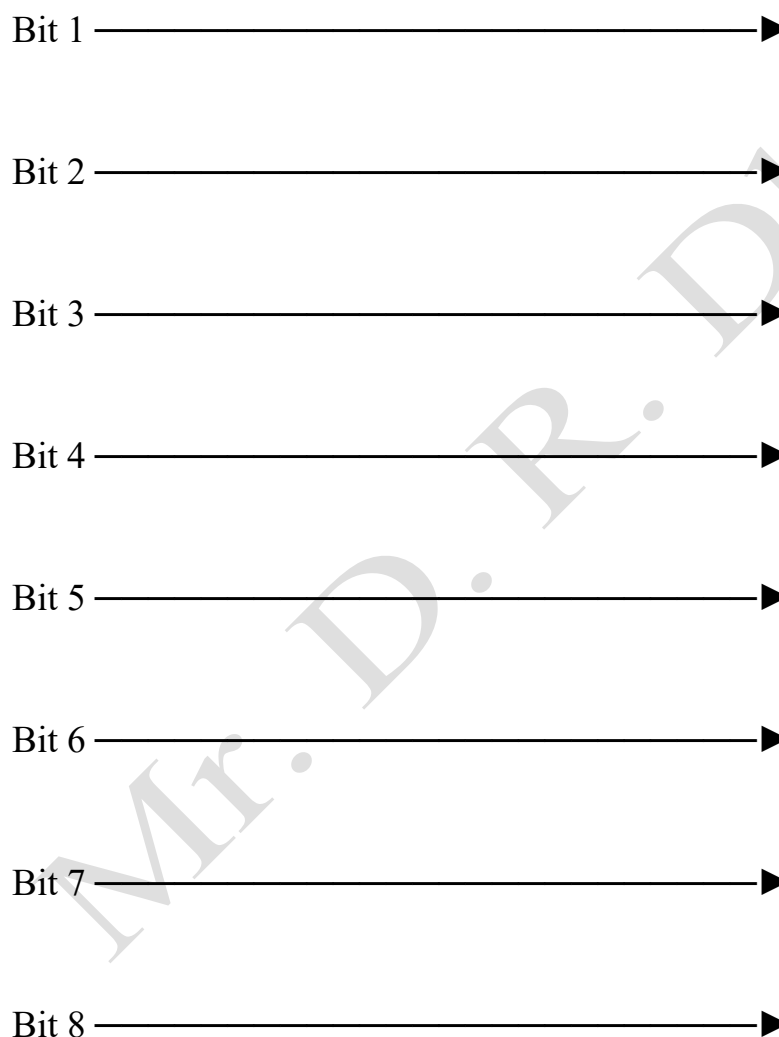
- Requires multiple communication lines.
- Installation cost is higher.
- Signal interference may occur between adjacent wires.
- Not suitable for long-distance communication.

Applications of Parallel Transmission

- Communication between a computer and a printer using the traditional parallel port.
- Internal communication between computer components.
- Short-distance data transfer between digital devices.

Parallel Transmission Diagram

Communication Lines



(All bits are transmitted simultaneously.)

Serial Transmission

Serial Transmission is a transmission mode in which data bits are transmitted one after another through a single communication line. Instead of sending multiple bits simultaneously, each bit is transmitted sequentially until the complete data is received.

Although only one bit is transmitted at a time, serial transmission is more reliable for long-distance communication. It requires fewer communication lines, reduces transmission cost, and minimizes signal interference. For these reasons, serial transmission is widely used in modern computer networks.

Characteristics of Serial Transmission

- Data is transmitted one bit at a time.
- Uses a single communication line.
- Suitable for long-distance communication.
- Less affected by signal interference.
- Reliable and economical.

Advantages of Serial Transmission

- Requires only one communication line.
- Lower installation and maintenance cost.
- Suitable for long-distance communication.
- Provides reliable data transmission.

Disadvantages of Serial Transmission

- Data is transmitted sequentially, one bit at a time.
- Transmission speed is lower than parallel transmission over short distances.
- Requires proper synchronization between communicating devices.

Applications of Serial Transmission

- Computer networks
- Internet communication
- USB communication
- Mobile communication
- Optical fiber communication
- Satellite communication

Serial Transmission Diagram

Communication Line

Bit 1 → Bit 2 → Bit 3 → Bit 4 → Bit 5 → Bit 6 → Bit 7 → Bit 8

(Bits are transmitted one after another through a single communication line.)

Difference Between Parallel Transmission and Serial Transmission

Parallel Transmission	Serial Transmission
Multiple bits are transmitted simultaneously.	One bit is transmitted at a time.
Uses multiple communication lines.	Uses a single communication line.
Provides higher speed over short distances.	Suitable for long-distance communication.
Installation cost is higher.	Installation cost is lower.
More susceptible to signal interference.	Less susceptible to signal interference.
Requires more cables and connectors.	Requires fewer cables and connectors.

Used mainly for short-distance communication.	Widely used in modern computer networks.
Example: Traditional parallel printer interface.	Examples: USB, Ethernet, Optical Fiber, Internet communication.

1.4 Data Transmission Media

Introduction

In a computer network, data is transmitted from one device to another in the form of signals. These signals require a path to travel between the sender and the receiver. This path is known as the **data transmission medium** or **communication medium**.

The transmission medium plays an important role in data communication because it affects the speed, distance, reliability, security, and cost of communication. The selection of an appropriate transmission medium depends on the communication requirements and the network environment.

Data Transmission Media

Data Transmission Media refers to the path through which data signals travel from the sender to the receiver.

Based on the method of data transmission, transmission media are classified into two types:

1. Guided (Wired) Transmission Media
2. Unguided (Wireless) Transmission Media

Guided (Wired) Transmission Media

Guided Transmission Media is a communication medium in which data signals travel through a physical path such as copper wires or optical fiber cables. Since the signals are guided through a fixed medium, guided transmission provides reliable communication, better security, and less interference. It is widely used in Local Area Networks (LANs), telephone networks, offices, educational institutions, and data centers.

The main types of guided transmission media are:

- Twisted Pair Cable
- Coaxial Cable
- Optical Fiber Cable

1. Twisted Pair Cable

A **Twisted Pair Cable** consists of two insulated copper wires twisted together. Twisting the wires helps reduce electromagnetic interference and signal noise, thereby improving communication quality. It is the most widely used transmission medium in computer networks because it is inexpensive, easy to install, and suitable for short-distance communication.

Twisted pair cables are commonly used in Ethernet-based LANs and telephone communication systems.

Characteristics

- Consists of two insulated copper wires twisted together.
- Low installation cost.
- Easy to install and maintain.
- Suitable for short-distance communication.
- Commonly used in LANs and telephone networks.

Advantages

- Low cost.
- Flexible and lightweight.
- Easy installation and maintenance.
- Widely available.

Disadvantages

- Limited transmission distance.
- Lower bandwidth than optical fiber.
- More susceptible to electromagnetic interference.

2. Coaxial Cable

A **Coaxial Cable** consists of a central copper conductor surrounded by an insulating layer, a metallic shield, and an outer protective covering. The metallic shield protects the signal from external interference, allowing more reliable data transmission than twisted pair cables.

Coaxial cables are widely used in cable television networks, broadband Internet connections, and CCTV systems.

Characteristics

- Contains a central copper conductor.
- Better protection against interference.
- Supports higher bandwidth than twisted pair cable.
- Suitable for medium-distance communication.

Advantages

- Better noise immunity.
- Higher bandwidth.
- Reliable signal transmission.
- Better performance than twisted pair cable.

Disadvantages

- More expensive than twisted pair cable.
- Thicker and less flexible.
- Installation is comparatively difficult.

3. Optical Fiber Cable

An **Optical Fiber Cable** transmits data in the form of light signals through very thin strands of glass. In some specialized applications, plastic optical fibers may also be used. Optical fiber provides extremely high transmission speed, large bandwidth, and excellent reliability with very low signal loss.

Due to these advantages, optical fiber is widely used in high-speed Internet services, telecommunication networks, backbone networks, and long-distance communication.

Characteristics

- Transmits data using light signals.
- Made of very thin strands of glass.
- Provides very high bandwidth.
- Suitable for long-distance communication.
- Immune to electromagnetic interference.

Advantages

- Extremely high transmission speed.
- Very high bandwidth.
- Long transmission distance.
- Secure communication.
- Minimal signal loss.

Disadvantages

- Higher installation cost.
- Installation and maintenance require specialized equipment.
- More difficult to repair than copper cables.

Unguided (Wireless) Transmission Media

Unguided Transmission Media is a communication medium in which data signals travel through free space without using physical cables. The signals are transmitted using electromagnetic waves. This type of transmission provides mobility and flexibility, making it suitable for wireless communication where cable installation is difficult or impractical.

The main types of unguided transmission media are:

- Radio Waves
- Microwaves
- Infrared Waves

1. Radio Waves

Radio Waves are electromagnetic waves capable of traveling over long distances. They can pass through buildings and other obstacles, making them suitable for many wireless communication systems.

Radio waves are commonly used in radio broadcasting, television broadcasting, Wireless LAN (Wi-Fi), and mobile communication systems.

2. Microwaves

Microwaves are high-frequency electromagnetic waves used for high-speed wireless communication. Microwave communication requires a **Line-of-Sight (LOS)** path between the transmitting and receiving antennas because these waves cannot easily pass through obstacles.

Microwaves are widely used in satellite communication, mobile networks, and point-to-point communication links.

3. Infrared Waves

Infrared Waves are electromagnetic waves used for short-distance wireless communication. They cannot penetrate walls or other solid objects and therefore are suitable only for communication between nearby devices.

Infrared waves are commonly used in television remote controls, wireless keyboards, wireless mice, and other short-range electronic devices.

Difference Between Guided and Unguided Transmission Media

Guided (Wired) Transmission Media	Unguided (Wireless) Transmission Media
Uses physical cables to transmit data.	Uses electromagnetic waves to transmit data.
Signals travel through a fixed physical path.	Signals travel through free space.
Provides better security and reliability.	Comparatively less secure.
Less affected by external interference.	More affected by environmental interference.
Suitable for stable and high-speed communication.	Suitable for mobile and wireless communication.
Installation requires physical cabling.	No physical cables are required.
Examples: Twisted Pair Cable, Coaxial Cable, Optical Fiber Cable.	Examples: Radio Waves, Microwaves, Infrared Waves.

1.5 Network Topologies – Bus, Star, Ring, Mesh

Introduction

In a computer network, computers and other networking devices must be connected in an organized manner to communicate with each other. The arrangement of these devices and the communication links between them determines how data travels through the network. This arrangement is known as **network topology**.

The choice of network topology affects the performance, reliability, scalability, maintenance, and cost of a network. Different topologies are suitable for different networking requirements depending on the size and purpose of the network.

Network Topology

A **Network Topology** is the physical or logical arrangement of computers, networking devices, communication links, and transmission media in a computer network. It defines how devices are connected to one another and how data is transmitted between them.

The commonly used network topologies are:

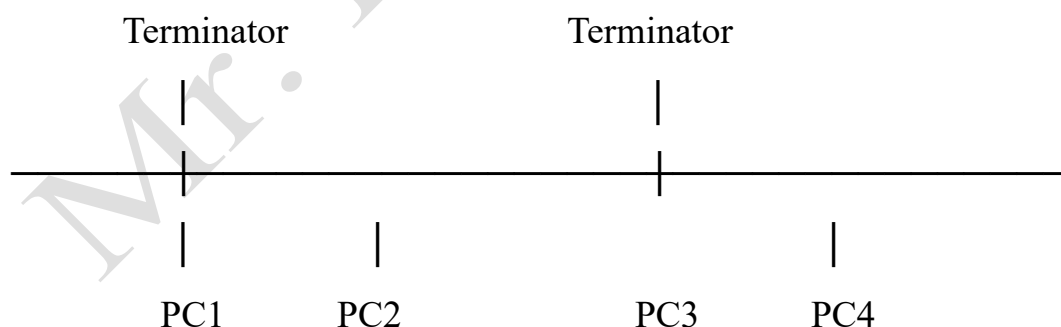
- Bus Topology
- Star Topology
- Ring Topology
- Mesh Topology

1. Bus Topology

A **Bus Topology** is a network topology in which all computers and networking devices are connected to a single communication cable called the **backbone cable**.

When a device transmits data, the signal travels through the backbone cable and reaches all connected devices. However, only the destination device accepts and processes the data, while the remaining devices ignore it. A **terminator** is installed at both ends of the backbone cable to absorb signals and prevent signal reflection.

Bus Topology Diagram



Advantages

- Simple network structure.
- Easy to install and maintain.
- Requires less cable.
- Low installation cost.

Disadvantages

- Failure of the backbone cable stops the entire network.
- Performance decreases as the number of connected devices increases.
- Difficult to identify faults.
- Limited network expansion.

Applications

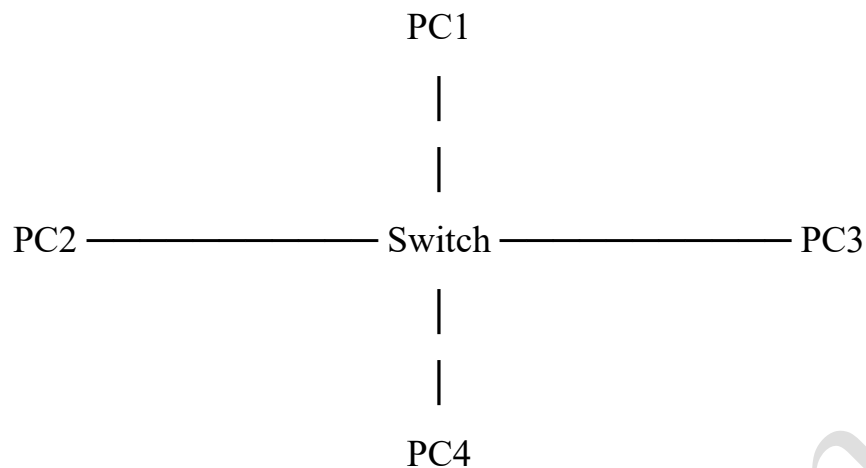
- Small Local Area Networks (LANs).
- Temporary network installations.
- Small office networks.

2. Star Topology

A **Star Topology** is a network topology in which every computer is connected individually to a central networking device such as a **switch** or **hub**. All communication between the connected devices passes through the central device.

If one computer or its communication link fails, the remaining computers continue to communicate normally. However, if the central switch or hub fails, the entire network becomes unavailable. In modern computer networks, a **switch** is preferred over a hub because it provides better performance and efficient data transmission.

Star Topology Diagram



Advantages

- Easy to install, manage, and expand.
- Easy to detect and isolate faults.
- Failure of one computer does not affect the remaining network.
- High network reliability.

Disadvantages

- Requires more cable than bus topology.
- Installation cost is higher.
- Failure of the central switch or hub stops the entire network.

Applications

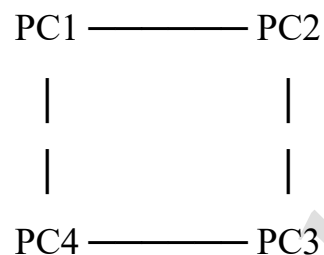
- Local Area Networks (LANs).
- Schools and colleges.
- Offices and business organizations.

3. Ring Topology

A **Ring Topology** is a network topology in which each computer is connected to two neighboring computers, forming a closed circular path. Data travels from one computer to the next until it reaches the destination.

In many ring networks, data transmission is controlled using a **token**. A device can transmit data only when it receives the token, thereby preventing data collisions and ensuring orderly communication.

Ring Topology Diagram



Advantages

- Orderly data transmission.
- Equal access for all connected devices.
- Better performance under moderate network traffic.
- Data collisions are minimized.

Disadvantages

- Failure of one device or communication link may interrupt the entire network.
- Adding or removing a device may temporarily disrupt communication.
- Troubleshooting is comparatively difficult.

Applications

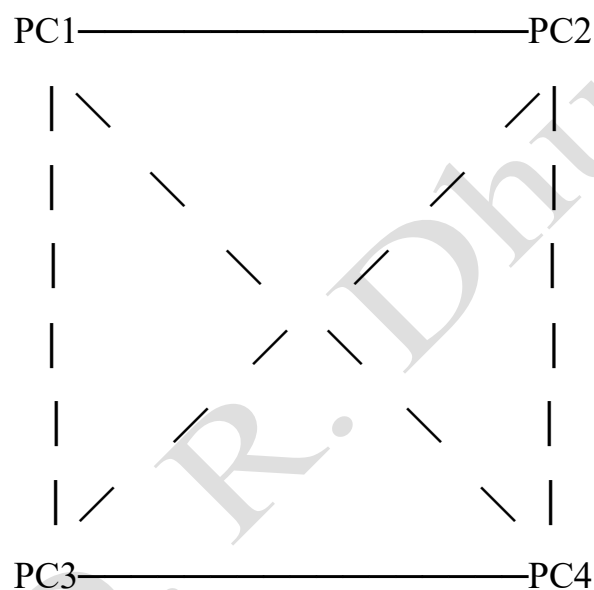
- Small office networks.
- Industrial communication systems.
- Token Ring networks.

4. Mesh Topology

A **Mesh Topology** is a network topology in which devices are interconnected through multiple communication links. In a **Full Mesh Topology**, every device is directly connected to every other device, whereas in a **Partial Mesh Topology**, only selected devices have multiple direct connections.

Multiple communication paths ensure that if one link fails, data can still reach its destination through another available path. This makes mesh topology highly reliable.

Mesh Topology Diagram



Advantages

- Very high reliability.
- Multiple communication paths.
- High fault tolerance.
- Easy fault isolation.
- Suitable for continuous communication.

Disadvantages

- Requires a large number of cables.
- High installation and maintenance cost.
- Complex network design.
- Difficult to expand.

Applications

- Data centers.
- Banking networks.
- Military communication systems.
- Telecommunication backbone networks.

Comparison of Network Topologies

Feature	Bus	Star	Ring	Mesh
Structure	Single backbone cable	Central switch or hub	Closed circular path	Multiple interconnections
Data Transmission	Shared backbone	Through central device	Around the ring	Multiple paths
Installation Cost	Low	Moderate	Moderate	High
Reliability	Low	High	Moderate	Very High
Fault Tolerance	Low	Moderate	Moderate	Very High
Cable Requirement	Least	More	Moderate	Maximum
Expansion	Limited	Easy	Difficult	Difficult

Common Use	Small LANs	Modern LANs	Specialized networks	Critical communication systems
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1.6 Network Types – LAN, MAN, WAN

Introduction

Computer networks are classified into different types based on the geographical area they cover and the distance over which they provide communication. Some networks connect devices within a single room or building, while others connect multiple cities, countries, or even continents.

The three most common types of computer networks are:

- Local Area Network (LAN)
- Metropolitan Area Network (MAN)
- Wide Area Network (WAN)

Network Types

Network Types are the classifications of computer networks based on the geographical area covered and the distance over which data communication takes place.

The three main network types are:

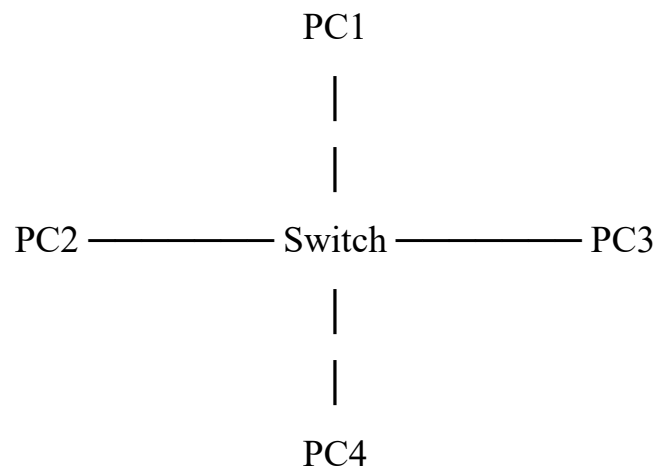
- Local Area Network (LAN)
- Metropolitan Area Network (MAN)
- Wide Area Network (WAN)

1. Local Area Network (LAN)

A **Local Area Network (LAN)** is a computer network that connects computers and other networking devices within a limited geographical area such as a room, office, building, school, college campus, laboratory, or factory.

A LAN provides high-speed communication and is usually owned, controlled, and maintained by a single organization. It enables users to share files, printers, applications, and Internet connections efficiently.

LAN Diagram



Characteristics

- Covers a small geographical area.
- Provides high-speed data communication.
- Owned and managed by a single organization.
- Easy to install and maintain.
- Supports resource sharing.

Advantages

- High data transmission speed.
- Low installation and maintenance cost.
- Easy sharing of files, printers, and Internet connection.
- Better security and reliability.

Disadvantages

- Limited geographical coverage.
- Difficult to expand over long distances.
- Failure of the central switch may affect network communication.

Applications

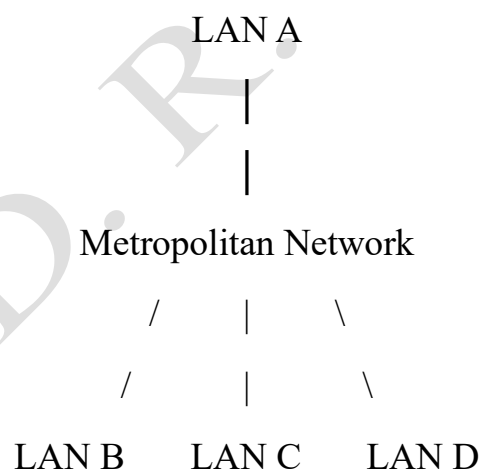
- Schools and colleges.
- Offices and business organizations.
- Computer laboratories.
- Libraries and hospitals.
- Small industries and factories.

2. Metropolitan Area Network (MAN)

A **Metropolitan Area Network (MAN)** is a computer network that connects multiple Local Area Networks (LANs) within a city or metropolitan area. It covers a larger geographical area than a LAN but a smaller area than a WAN.

A MAN is commonly used by government organizations, banks, cable television providers, educational institutions, and large business organizations to provide communication between different locations within the same city.

MAN Diagram



Characteristics

- Covers a city or metropolitan area.
- Connects multiple LANs.
- Provides high-speed communication.
- Larger than a LAN but smaller than a WAN.
- May be privately or publicly managed.

Advantages

- Connects multiple LANs efficiently.
- Supports communication across an entire city.
- High communication speed.
- Enables resource sharing between different locations.

Disadvantages

- Higher installation cost than a LAN.
- More complex network management.
- Maintenance requires skilled professionals.

Applications

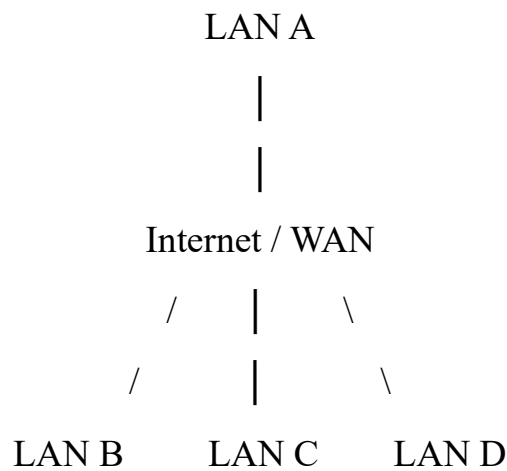
- City-wide banking networks.
- Government departments.
- Cable television networks.
- Large business organizations with multiple city branches.

3. Wide Area Network (WAN)

A **Wide Area Network (WAN)** is a computer network that connects computers and networks over very large geographical areas such as countries or continents. It interconnects multiple LANs and MANs using communication links such as leased lines, fiber optic cables, satellite links, and the Internet.

The **Internet** is the largest and most widely used example of a WAN.

WAN Diagram



Characteristics

- Covers very large geographical areas.
- Connects multiple LANs and MANs.
- Suitable for long-distance communication.
- Uses public and private communication networks.
- Provides worldwide connectivity.

Advantages

- Enables communication over long distances.
- Connects offices located in different cities and countries.
- Supports global information sharing.
- Suitable for multinational organizations.

Disadvantages

- High installation and maintenance cost.
- Lower communication speed than a LAN.
- Complex network management.
- Greater dependence on communication service providers.

Applications

- The Internet.
- Multinational companies.
- Banking networks.
- Airline reservation systems.
- International business communication.

Difference Between LAN, MAN, and WAN

Feature	LAN	MAN	WAN
Full Form	Local Area Network	Metropolitan Area Network	Wide Area Network
Coverage Area	Room, building, or campus	City or metropolitan area	Countries or continents
Typical Distance	Up to a few kilometers	Covers an entire city	Covers countries or continents
Network Size	Small	Medium	Large
Data Transmission Speed	High	High to Moderate	Moderate
Ownership	Single organization	Private or public organization	Multiple organizations and service providers
Installation Cost	Low	Moderate	High
Maintenance	Easy	Moderate	Complex
Common Examples	School, office, laboratory	City-wide banking network, cable television network	Internet, multinational company network

Module No. 2

Network Architecture and IP Address

2.1 Network Standards, Ethernet, Types of Ethernet

Introduction

A computer network consists of devices manufactured by different companies. For these devices to communicate efficiently, they must follow a common set of rules and technical specifications. These rules are known as **network standards**. Network standards ensure compatibility, interoperability, reliability, and efficient communication among networking devices.

One of the most widely used networking technologies based on international standards is **Ethernet**. Ethernet is the standard technology used for building Local Area Networks (LANs). It provides reliable, high-speed communication between computers and networking devices and has become the most widely used wired LAN technology worldwide.

Network Standards

Network Standards are a set of rules, guidelines, and technical specifications that define how networking devices communicate and exchange data in a computer network.

They ensure that networking devices manufactured by different vendors can communicate with one another without compatibility issues.

Importance of Network Standards

- Ensure compatibility between networking devices.
- Enable communication among devices from different manufacturers.
- Improve network reliability and performance.
- Simplify network design and maintenance.
- Support the development of scalable and standardized networks.

IEEE 802 Standards

Among the various organizations that develop networking standards, the **Institute of Electrical and Electronics Engineers (IEEE)** is one of the most important. The IEEE has developed the **IEEE 802** family of standards for Local Area Networks (LANs) and Metropolitan Area Networks (MANs).

Some important IEEE 802 standards are given below:

IEEE Standard	Network Technology
IEEE 802.3	Ethernet (Wired LAN)
IEEE 802.11	Wireless LAN (Wi-Fi)
IEEE 802.15	Bluetooth (WPAN)
IEEE 802.16	WiMAX (Wireless MAN)

Ethernet

Ethernet is the most widely used wired Local Area Network (LAN) technology for connecting computers and networking devices. It follows the **IEEE 802.3** standard and enables reliable communication using Ethernet cables and network switches.

Ethernet provides high-speed, reliable, and cost-effective communication. It is widely used in homes, schools, colleges, offices, industries, and data centers. Ethernet communication uses **MAC (Media Access Control) addresses** to identify devices and deliver data to the correct destination.

Basic Working of Ethernet

During Ethernet communication, data is transmitted in the form of **Ethernet frames**.

The basic working of Ethernet is as follows:

1. A computer prepares the data for transmission.
2. The data is divided into small units called **Ethernet frames**.
3. The Ethernet frames are transmitted through an Ethernet cable to a network switch.
4. The switch reads the destination MAC address and forwards the frames to the appropriate device.
5. The destination device receives and processes the data.

Characteristics of Ethernet

- Based on the IEEE 802.3 standard.
- Uses MAC addresses for device identification.
- Provides reliable wired communication.
- Supports high-speed data transmission.
- Easy to install and maintain.
- Cost-effective networking technology.
- Widely used in Local Area Networks (LANs).

Advantages

- High data transmission speed.
- Reliable communication.
- Low installation and maintenance cost.
- Easy network expansion.
- Supports communication between devices from different manufacturers.

Disadvantages

- Limited mobility because physical cables are required.
- Cable installation may be difficult in large buildings.
- Cable damage may interrupt communication.
- Expansion requires additional cabling.

Types of Ethernet

Ethernet technology has evolved over time to provide higher communication speeds. Based on the maximum data transmission speed, Ethernet is classified into the following types.

Evolution of Ethernet

Standard Ethernet



10 Mbps



Fast Ethernet

• 100 Mbps



Gigabit Ethernet

1 Gbps



10 Gigabit Ethernet

10 Gbps

1. Standard Ethernet

Standard Ethernet is the original Ethernet technology that supports a maximum data transmission speed of **10 Mbps (Megabits per second)**. It is suitable for small networks with basic communication requirements.

Characteristics

- Speed: 10 Mbps
- IEEE Standard: IEEE 802.3
- Suitable for small networks
- Low bandwidth compared to modern Ethernet

2. Fast Ethernet

Fast Ethernet supports a maximum data transmission speed of **100 Mbps**. It provides ten times higher speed than Standard Ethernet and is widely used in small and medium-sized Local Area Networks.

Characteristics

- Speed: 100 Mbps
- IEEE Standard: IEEE 802.3u
- Faster than Standard Ethernet
- Suitable for office and educational networks

3. Gigabit Ethernet

Gigabit Ethernet supports a maximum data transmission speed of **1 Gbps (1000 Mbps)**. It is commonly used in modern organizations, educational institutions, and enterprise networks where high-speed communication is required. It is also suitable for multimedia applications and large file transfers.

Characteristics

- Speed: 1 Gbps
- IEEE Standard: IEEE 802.3ab / IEEE 802.3z
- High bandwidth
- Suitable for high-speed LANs

4. 10 Gigabit Ethernet

10 Gigabit Ethernet supports a maximum data transmission speed of **10 Gbps**. It is mainly used in enterprise networks, backbone networks, servers, and data centers where very high-speed communication is required.

Characteristics

- Speed: 10 Gbps
- IEEE Standard: IEEE 802.3ae
- Very high bandwidth
- Suitable for enterprise networks and data centers

Difference Between Types of Ethernet

Feature	Standard Ethernet	Fast Ethernet	Gigabit Ethernet	10 Gigabit Ethernet
Maximum Speed	10 Mbps	100 Mbps	1 Gbps	10 Gbps
IEEE Standard	IEEE 802.3	IEEE 802.3u	IEEE 802.3ab / IEEE 802.3z	IEEE 802.3ae
Performance	Low	Moderate	High	Very High
Typical Applications	Small LANs	Offices and Schools	Organizations, Campuses, Multimedia Applications	Data Centers, Enterprise and Backbone Networks
Cost	Low	Moderate	High	Very High

2.2 Client and Server Architecture

Introduction

In a computer network, devices communicate with each other to share data, resources, and services. Some computers request services, while others provide those services. This method of communication is known as **Client-Server Architecture**.

Client-Server Architecture is one of the most widely used network architectures because it provides centralized management, efficient resource sharing, better security, and easy maintenance. It is commonly used in organizations, educational institutions, banks, hospitals, e-commerce websites, cloud services, and Internet-based applications.

Client-Server Architecture

Client-Server Architecture is a network architecture in which one or more computers, called **servers**, provide resources or services, while other computers, called **clients**, request and use those resources or services through a network.

The server manages shared resources, processes client requests, and sends the requested information or services back to the clients.

Components of Client-Server Architecture

1. Client

A **Client** is a computer or device that requests services or resources from a server. A client communicates with the server using application software such as a web browser, email client, database application, or mobile application.

Characteristics

- Requests services or resources from the server.
- Initiates communication with the server.
- Uses the services provided by the server.
- Can be a desktop computer, laptop, smartphone, or tablet.
- Depends on the server for shared resources.

2. Server

A **Server** is a powerful computer or system that provides services, resources, or data to multiple clients connected through a network. It receives client requests, processes them, and sends the appropriate response.

A server is designed to operate continuously and efficiently while handling requests from many clients simultaneously.

Characteristics

- Provides services to multiple clients.
- Stores shared files, databases, and applications.
- Processes client requests.
- Operates continuously to provide uninterrupted services.
- Offers centralized management and security.
- Supports multiple users simultaneously.

Working of Client-Server Architecture

The working of Client-Server Architecture is as follows:

1. A client sends a request to the server through the network.
2. The server receives and processes the request.
3. The server accesses the required data or resource.
4. The server prepares the response.
5. The server sends the requested information or service back to the client.
6. The client receives the response and displays or uses the requested data.

Advantages

- Centralized management of data and resources.
- Better security through centralized authentication and access control.
- Easy backup and recovery of data.
- Efficient resource sharing among multiple users.
- Easy network administration and maintenance.
- Supports multiple clients simultaneously.
- Easy to upgrade and expand the network.

Disadvantages

- Failure of the server may interrupt network services.
- Higher installation and maintenance cost.
- Requires a dedicated server computer.
- Network performance depends on the server's capacity.
- Requires skilled administrators for server management.

Applications

- Banking systems.
- Educational institutions.
- Hospital management systems.
- E-commerce websites.
- Email services.
- Database management systems.
- Cloud computing services.
- Online reservation systems.

Difference Between Client and Server

Feature	Client	Server
Definition	Requests services or resources from a server.	Provides services or resources to clients.
Role	Initiates communication by sending requests.	Receives, processes, and responds to client requests.
Function	Uses shared resources and services.	Manages shared resources and services.
Processing Capability	Usually lower.	Usually higher.
Resource Management	Does not manage shared resources.	Manages and controls shared resources.
Number in a Network	Many clients can exist.	One or more servers can serve many clients.
Examples	Desktop, Laptop, Smartphone, Tablet	Web Server, File Server, Database Server, Mail Server

Client-Server Architecture vs Peer-to-Peer Architecture

Feature	Client-Server Architecture	Peer-to-Peer (P2P) Architecture
Central Server	Required	Not Required
Resource Management	Centralized	Decentralized
Security	Better	Comparatively Lower
Administration	Easy	Difficult
Cost	Higher	Lower
Suitable For	Medium and Large Networks	Small Networks
Examples	Banking System, College Network, Web Applications	Home Networks, Small Office Networks, File Sharing

Note: In Client-Server Architecture, a dedicated server provides services to multiple clients, whereas in Peer-to-Peer Architecture, every computer can act as both a client and a server.

2.3 Internet versus Intranet

Introduction

Computer networks have transformed the way people communicate, access information, and share resources. Two commonly used networking environments are the **Internet** and the **Intranet**. Although both use similar networking technologies and communication protocols, they differ significantly in terms of accessibility, ownership, security, and purpose.

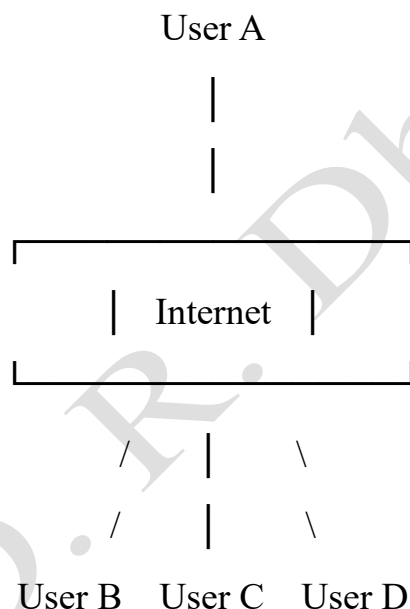
Understanding the differences between the Internet and the Intranet helps individuals and organizations choose the appropriate network for communication, collaboration, and information sharing.

Internet

The **Internet** is a worldwide network of interconnected computer networks that enables millions of users to communicate and share information across the globe. It is a **public network** that uses the **TCP/IP (Transmission Control Protocol/Internet Protocol)** suite for communication between connected devices.

The Internet provides a wide range of services such as web browsing, email, online banking, e-commerce, cloud computing, video conferencing, social networking, and online education.

Internet Diagram



Characteristics

- Public network accessible worldwide.
- Connects millions of computers and networks.
- Uses TCP/IP communication protocols.
- Supports a wide variety of online services.
- No single organization owns or controls the Internet.

Advantages

- Enables worldwide communication.
- Provides quick access to information and knowledge.
- Supports online education, banking, shopping, and entertainment.
- Facilitates cloud computing and remote collaboration.
- Connects users across different countries.

Disadvantages

- Greater security risks.
- Privacy concerns due to public accessibility.
- Vulnerable to cyber attacks, malware, and phishing.
- Internet speed depends on the service provider and network traffic.

Applications

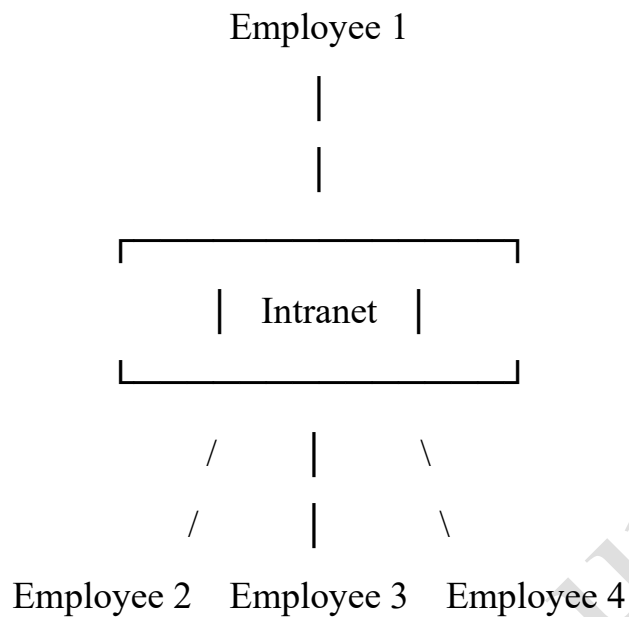
- Web browsing.
- Email communication.
- Online banking.
- E-commerce.
- Social networking.
- Video conferencing.
- Cloud computing.
- Online learning.

Intranet

An **Intranet** is a private computer network that uses Internet technologies and the **TCP/IP** protocol suite to share information, resources, and services within an organization. Access to an intranet is restricted to authorized users such as employees, faculty members, or students.

An intranet is owned and managed by a single organization and is primarily used for internal communication, document sharing, employee services, and organizational applications.

Intranet Diagram



Characteristics

- Private network.
- Accessible only to authorized users.
- Owned and managed by a single organization.
- Uses Internet technologies and TCP/IP protocols.
- Provides secure internal communication.
- Supports centralized information sharing.

Advantages

- High level of security.
- Fast communication within the organization.
- Easy sharing of documents and organizational resources.
- Improves collaboration among employees.
- Reduces communication and operational costs.

Disadvantages

- Accessible only to authorized users.
- Installation and maintenance costs are comparatively high.
- Requires continuous administration and technical support.
- Limited to the organization's network.

Applications

- Employee portals.
- Human Resource (HR) management systems.
- Internal email systems.
- Document management systems.
- Internal project management.
- College or university portals.
- Organizational announcements.

Difference Between Internet and Intranet

Feature	Internet	Intranet
Definition	A worldwide public network connecting millions of computers and networks.	A private network used within an organization.
Accessibility	Accessible to anyone with an Internet connection.	Accessible only to authorized users.
Ownership	No single organization owns or manages it.	Owned and managed by a single organization.
Network Type	Public network.	Private network.
Security	Comparatively less secure.	More secure due to restricted access.

Users	General public worldwide.	Employees, students, faculty, or authorized members.
Coverage Area	Worldwide.	Limited to an organization or institution.
Purpose	Global communication and information sharing.	Internal communication and resource sharing.
Data Accessibility	Publicly available information.	Internal organizational information.
Examples	Google Search, Wikipedia, Gmail, Online Banking	Company Employee Portal, College Intranet, HR Portal, Internal Document Management System

Key Differences

- The **Internet** is a public network available worldwide, whereas an **Intranet** is a private network accessible only to authorized users.
- The Internet is used for global communication and public information sharing, whereas an Intranet is used for secure communication and information sharing within an organization.
- The Internet has no single owner, whereas an Intranet is owned and managed by an individual organization.
- The Internet can be accessed from anywhere with an Internet connection, whereas an Intranet is generally accessible only within the organization's network or through secure remote access.
- An Intranet provides better security than the Internet because access is restricted to authorized users and organizational security policies.

2.4 Connection-Oriented and Connectionless Services

Introduction

In a computer network, data can be transmitted between devices using different communication methods. Depending on how communication is established between the sender and the receiver, network services are classified into **Connection-Oriented Services** and **Connectionless Services**.

These services determine how data is transferred across a network and influence the reliability, speed, and efficiency of communication. Understanding these services is essential for selecting the appropriate method of data transmission in different networking applications.

Connection-Oriented Service

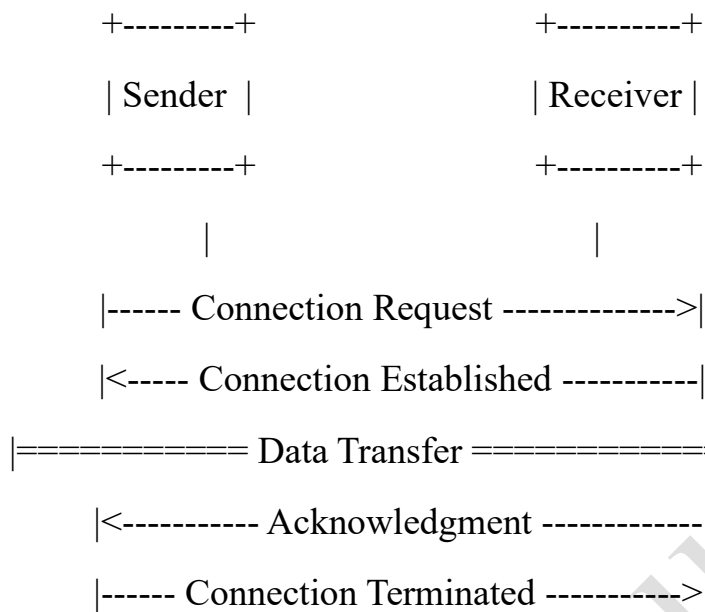
A **Connection-Oriented Service** is a communication service in which a connection is established between the sender and the receiver before data transmission begins. The connection remains active until all data has been transmitted successfully and is then terminated.

This service provides reliable communication by ensuring that data is delivered accurately and in the correct sequence. The **Transmission Control Protocol (TCP)** is the most common example of a connection-oriented protocol.

Working of Connection-Oriented Service

1. The sender sends a connection request to the receiver.
2. The receiver accepts the request, and a connection is established.
3. Data is transmitted in the correct sequence.
4. The receiver acknowledges the received data.
5. After successful communication, the connection is terminated.

Connection-Oriented Service Diagram



Characteristics

- Connection is established before data transmission.
- Reliable communication.
- Data is delivered in the correct sequence.
- Uses acknowledgments and error checking.
- Suitable for applications requiring accurate and reliable data delivery.

Advantages

- Reliable data transmission.
- Ensures data reaches the destination.
- Maintains the correct order of data packets.
- Supports error detection and recovery.

Disadvantages

- Requires additional time to establish the connection.
- Higher communication overhead.
- Slower than connectionless communication.

Applications

- TCP communication.
- File Transfer Protocol (FTP).
- Hypertext Transfer Protocol (HTTP/HTTPS).
- Remote login services (SSH/Telnet).
- Online banking and database transactions.

Connectionless Service

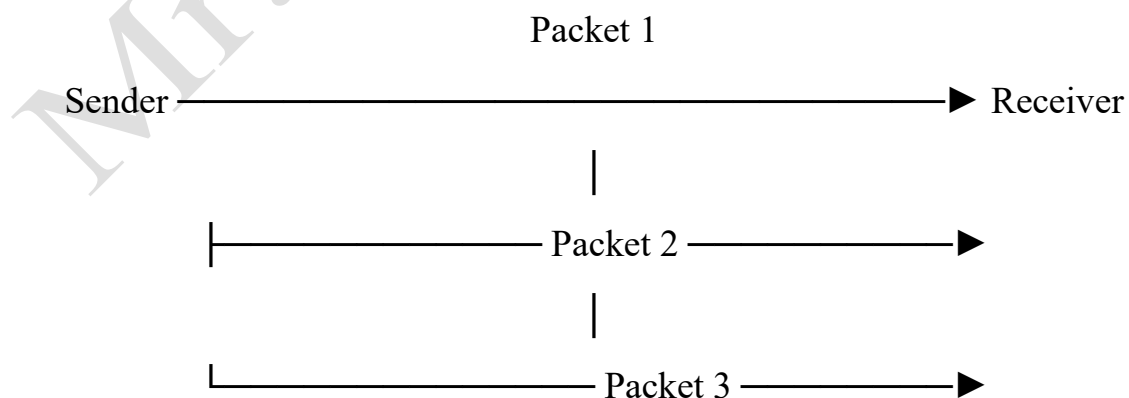
A **Connectionless Service** is a communication service in which data is transmitted without establishing a dedicated connection between the sender and the receiver. Each data packet is transmitted independently and may travel through different network paths before reaching its destination.

This service provides faster communication but does not guarantee the delivery, sequence, or reliability of data. The **User Datagram Protocol (UDP)** is the most common example of a connectionless protocol.

Working of Connectionless Service

1. The sender divides the data into packets.
2. Each packet is transmitted independently.
3. Different packets may travel through different network paths.
4. The receiver accepts the packets without prior connection establishment.

Connectionless Service Diagram



Characteristics

- No connection establishment is required.
- Data packets are transmitted independently.
- Faster communication.
- No acknowledgment is required.
- Does not guarantee packet delivery or sequence.
- Suitable for real-time communication.

Advantages

- Faster data transmission.
- Low communication overhead.
- Efficient for real-time applications.
- Suitable for broadcasting and multicasting.

Disadvantages

- No guarantee of data delivery.
- Packets may arrive out of sequence.
- No built-in error recovery.
- Less reliable than connection-oriented communication.

Applications

- UDP communication.
- Domain Name System (DNS).
- Voice over IP (VoIP).
- Live video streaming.
- Online gaming.
- Video conferencing.

Difference Between Connection-Oriented and Connectionless Services

Feature	Connection-Oriented Service	Connectionless Service
Connection	Required before data transmission	Not required
Reliability	High	Lower
Data Delivery	Guaranteed	Not guaranteed
Packet Sequence	Maintained	May not be maintained
Acknowledgment	Required	Not required
Error Recovery	Available	Not available
Communication Speed	Comparatively slower	Faster
Common Protocols / Applications	TCP, FTP, HTTP/HTTPS, SSH	UDP, DNS, VoIP, Online Gaming

2.5 IP Address Classes

Introduction

Every device connected to a computer network must have a unique address for identification and communication. This address is known as an **IP (Internet Protocol) Address**. An IP address enables devices to send and receive data over a network or the Internet.

In **IPv4**, IP addresses are classified into different classes based on the size of the network and the number of hosts they can support. This classification helps in the efficient allocation and management of IP addresses in computer networks.

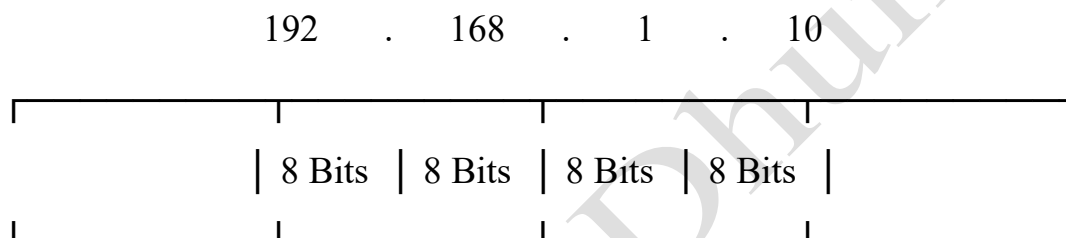
IP Address

An **IP (Internet Protocol) Address** is a unique numerical address assigned to every device connected to a computer network. It uniquely identifies a device on the network and enables data packets to be routed correctly from the source to the destination.

An IPv4 address consists of **32 bits** and is represented as **four decimal numbers separated by dots (.)**. Each decimal number is called an **octet** and ranges from **0 to 255**.

Example: 192.168.1.10

Structure of an IPv4 Address



Total = 32 Bits (4 Octets)

IP Address Classes

IPv4 addresses are divided into five classes based on the value of the first octet and the size of the network they support. These classes are **Class A**, **Class B**, **Class C**, **Class D**, and **Class E**.

Classes **A**, **B**, and **C** are used for assigning IP addresses to hosts in networks, whereas **Class D** is used for multicast communication and **Class E** is reserved for experimental purposes.

1. Class A

Class A addresses are designed for very large networks that require a large number of host addresses.

Characteristics

- First Octet Range: **1 – 126**
- Leading Bit Pattern: **0**
- Default Subnet Mask: **255.0.0.0**
- Network ID: First Octet
- Host ID: Last Three Octets

Example: 25.100.50.10

2. Class B

Class B addresses are designed for medium-sized networks.

Characteristics

- First Octet Range: **128 – 191**
- Leading Bit Pattern: **10**
- Default Subnet Mask: **255.255.0.0**
- Network ID: First Two Octets
- Host ID: Last Two Octets

Example: 172.16.25.100

3. Class C

Class C addresses are designed for small networks with a limited number of hosts.

Characteristics

- First Octet Range: **192 – 223**
- Leading Bit Pattern: **110**
- Default Subnet Mask: **255.255.255.0**
- Network ID: First Three Octets
- Host ID: Last Octet

Example: 192.168.1.50

4. Class D

Class D addresses are reserved for **multicast communication**, in which data is transmitted from one sender to multiple selected receivers simultaneously.

Characteristics

- First Octet Range: **224 – 239**
- Leading Bit Pattern: **1110**
- Used for multicast communication.
- Not divided into Network ID and Host ID.

Example: 230.10.20.30

5. Class E

Class E addresses are reserved for experimental and research purposes. They are not used for normal communication in computer networks.

Characteristics

- First Octet Range: **240 – 255**
 - Leading Bit Pattern: **1111**
 - Reserved for experimental purposes.
 - Not divided into Network ID and Host ID.
- Example:** 245.20.30.40

Difference Between IP Address Classes

Feature	Class A	Class B	Class C	Class D	Class E
First Octet Range	1–126	128–191	192–223	224–239	240–255
Leading Bit Pattern	0	10	110	1110	1111
Default Subnet Mask	255.0.0.0	255.255.0.0	255.255.255.0	Not Applicable	Not Applicable
Network ID	1 Octet	2 Octets	3 Octets	Not Applicable	Not Applicable
Host ID	3 Octets	2 Octets	1 Octet	Not Applicable	Not Applicable
Network Size	Very Large	Medium	Small	Multicast	Experimental
Primary Use	Host Addressing	Host Addressing	Host Addressing	Multicast Communication	Experimental Purposes

2.6 IPv4 vs IPv6

Introduction

The rapid growth of the Internet and the increasing number of connected devices has created a demand for more IP addresses. **IPv4 (Internet Protocol Version 4)** has been the standard Internet Protocol for many years, but its limited address space led to the development of **IPv6 (Internet Protocol Version 6)**.

IPv6 was introduced to overcome the limitations of IPv4 by providing a much larger address space, improved routing efficiency, enhanced security features, and better support for modern Internet applications. Today, both IPv4 and IPv6 are widely used in computer networks.

IPv4 (Internet Protocol Version 4)

IPv4 is the fourth version of the Internet Protocol and is the most widely used protocol for identifying devices on a network. It uses a **32-bit address**, allowing approximately **4.3 billion unique IP addresses**.

An IPv4 address is represented in **dotted-decimal notation**, consisting of four decimal numbers separated by dots (.).

Example: 192.168.1.10

Characteristics of IPv4

- Uses a 32-bit address.
- Provides approximately 4.3 billion unique IP addresses.
- Uses dotted-decimal notation.
- Supports manual and dynamic address configuration.
- Widely used in existing computer networks.
- The limited address space can lead to address exhaustion as the number of Internet-connected devices increases.

IPv6 (Internet Protocol Version 6)

IPv6 is the latest version of the Internet Protocol developed to overcome the limitations of IPv4, particularly the shortage of available IP addresses. It uses a **128-bit address**, providing an extremely large address space for present and future networking requirements.

An IPv6 address is represented in **hexadecimal notation**, consisting of eight groups separated by colons (:).

Example: 2001:0db8:85a3:0000:0000:8a2e:0370:7334

Characteristics of IPv6

- Uses a 128-bit address.
- Provides an extremely large address space.
- Uses hexadecimal notation.
- Supports automatic address configuration.
- Offers improved routing efficiency.
- Provides native support for IPsec security.
- Designed to support modern Internet and IoT applications.

Advantages of IPv6 over IPv4

- Provides a significantly larger address space.
- Eliminates the problem of IPv4 address exhaustion.
- Supports more efficient packet routing.
- Provides native support for IPsec security.
- Reduces the need for Network Address Translation (NAT).
- Supports automatic address configuration.
- Supports the growing number of Internet-connected and IoT devices.
- Designed to meet future networking requirements.

Difference Between IPv4 and IPv6

Feature	IPv4	IPv6
Full Name	Internet Protocol Version 4	Internet Protocol Version 6
Address Length	32 Bits	128 Bits
Address Format	Dotted Decimal	Hexadecimal
Address Example	192.168.1.10	2001:0db8:85a3:0000:0000:8a2e:0370:7334
Address Space	Approximately 4.3 Billion Addresses	Approximately 3.4×10^{38} Addresses
Header Size	Variable (20–60 Bytes)	Fixed (40 Bytes)
Address Configuration	Manual or DHCP	Manual, DHCPv6, or Stateless Address Auto Configuration (SLAAC)
Broadcast Communication	Supported	Not Supported (Uses Multicast Instead)
Multicast Support	Supported	Supported
Anycast Support	Not Supported	Supported
Security	Optional IPsec Support	Native Support for IPsec

Network Address Translation (NAT)	Commonly Used	Generally Not Required
Routing	Less Efficient	More Efficient
Scalability	Limited by Address Space	Highly Scalable
Address Exhaustion	Possible	Not a Practical Concern

Important points

1. IPv4 uses a 32-bit address, whereas IPv6 uses a 128-bit address, providing a much larger address space.
2. IPv4 addresses are written in dotted-decimal notation, while IPv6 addresses are written in hexadecimal notation separated by colons.
3. IPv6 was developed to overcome the address exhaustion problem of IPv4 and to support the growing number of Internet-connected devices.
4. IPv6 provides native support for IPsec security, automatic address configuration, and more efficient routing than IPv4.
5. Although IPv4 is still widely used, IPv6 is the future Internet Protocol designed to meet modern and future networking requirements.

Module No. 3

Protocols and Network Models

3.1 Network Protocols: TCP/IP and SMTP

Introduction

A computer network allows multiple devices to communicate and exchange information. For successful communication, all devices must follow a common set of rules and standards. These rules are known as **network protocols**. Network protocols define how data is formatted, transmitted, received, and interpreted, ensuring reliable and efficient communication between devices.

Among the various network protocols used in computer networks, **TCP/IP** and **SMTP** are two of the most important. TCP/IP is the standard protocol suite used for Internet communication, while SMTP is the standard protocol used for sending e-mails.

Network Protocol

A **Network Protocol** is a set of rules and standards that governs communication between devices connected to a computer network. It specifies how data is transmitted, received, addressed, and processed so that different devices can communicate accurately and efficiently.

Characteristics of Network Protocols

- Define rules for communication between network devices.
- Ensure reliable and accurate data transmission.
- Support error detection and recovery.
- Enable interoperability among different hardware and software platforms.
- Standardize communication across computer networks.

TCP/IP (Transmission Control Protocol / Internet Protocol)

TCP/IP is a **suite of communication protocols** used for communication over the Internet and other computer networks. The two primary protocols in this suite are **Transmission Control Protocol (TCP)** and **Internet Protocol (IP)**.

- **TCP** is responsible for reliable data transmission by dividing data into packets, ensuring error checking, sequencing, and retransmission of lost packets.
- **IP** is responsible for assigning logical addresses and routing packets from the source device to the destination device across the network.

Together, TCP and IP provide reliable end-to-end communication between networked devices.

Working of TCP/IP

1. The sender divides the data into smaller packets using TCP.
2. Each packet is encapsulated with the required TCP and IP information, including the source and destination IP addresses.
3. IP routes the packets through the network to the destination.
4. TCP verifies that all packets have been received correctly.
5. The packets are reassembled into the original data at the destination.

Features of TCP/IP

- Provides reliable end-to-end communication.
- Supports packet-based data transmission.
- Performs error detection and recovery.
- Supports communication between heterogeneous networks.
- Highly scalable and widely used on the Internet.

Advantages

- Reliable communication.
- Accurate delivery of data packets.
- Supports large-scale computer networks.
- Compatible with different hardware and operating systems.
- Flexible and scalable protocol suite.

Applications

- Web browsing
- File transfer
- Internet communication
- Remote login
- E-mail communication
- Online services

SMTP (Simple Mail Transfer Protocol)

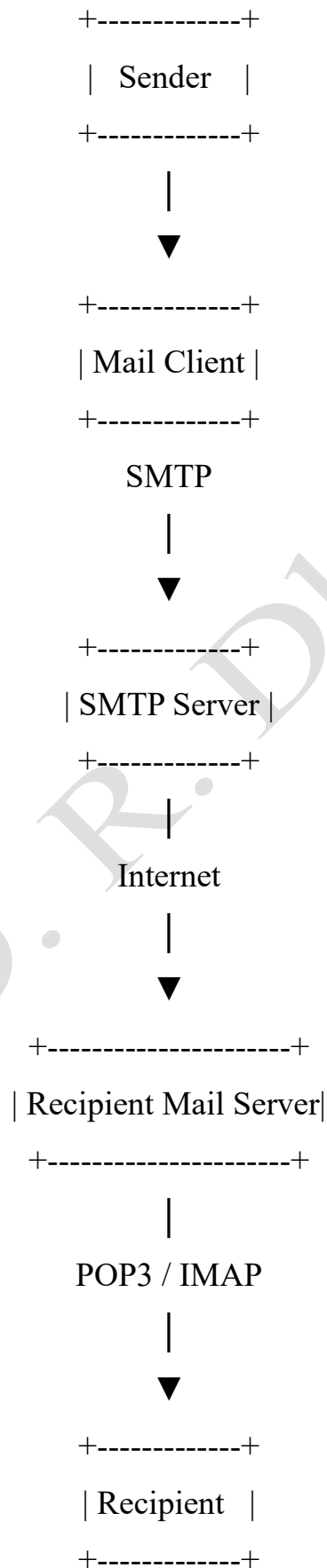
SMTP (Simple Mail Transfer Protocol) is a standard **Application Layer** protocol used for sending e-mails over the Internet. It transfers outgoing e-mail messages from the sender's mail client to the sender's mail server and between mail servers until the message reaches the recipient's mail server.

SMTP is used only for sending e-mails. The recipient retrieves the e-mail using protocols such as **POP3** or **IMAP**.

Working of SMTP

1. The sender composes an e-mail message.
2. The mail client sends the message to the sender's SMTP server.
3. The SMTP server forwards the message through the Internet to the recipient's mail server.
4. The recipient retrieves the e-mail using POP3 or IMAP.

SMTP Communication



Features of SMTP

- Standard protocol for sending e-mails.
- Operates at the Application Layer.
- Transfers e-mails between mail servers.
- Uses TCP for reliable message delivery.
- Supported by almost all e-mail service providers.

Advantages

- Fast and reliable e-mail transmission.
- Easy to implement and use.
- Supports communication between different mail servers.
- Compatible with most e-mail applications.
- Efficient for personal and business communication.

Applications

- Personal e-mail communication.
- Corporate e-mail systems.
- Banking and financial notifications.
- E-commerce order confirmations.
- Educational and institutional e-mail services.
- Automated notification systems.

Difference Between TCP/IP and SMTP

Feature	TCP/IP	SMTP
Full Form	Transmission Control Protocol / Internet Protocol	Simple Mail Transfer Protocol
Protocol Type	Protocol Suite	Application Layer Protocol
Purpose	Enables communication and data transfer over computer networks	Sends e-mail messages over the Internet
Main Function	Provides reliable communication, addressing, and routing	Transfers outgoing e-mail messages
Data Type	All types of network data	E-mail messages only
Reliability	Provides reliable communication using TCP	Uses TCP for reliable e-mail delivery
Common Applications	Web browsing, file transfer, Internet communication, remote login	Personal, business, and automated e-mail communication

3.2 DHCP and DNS

Introduction

In a computer network, every device requires an **IP address** to communicate with other devices. Similarly, users access websites using easy-to-remember **domain names** instead of numerical IP addresses. Two important network services that make these tasks possible are **DHCP (Dynamic Host Configuration Protocol)** and **DNS (Domain Name System)**.

DHCP automatically assigns IP addresses and other network configuration parameters to devices, while DNS translates domain names into IP addresses. Together, these protocols simplify network management and Internet communication.

DHCP (Dynamic Host Configuration Protocol)

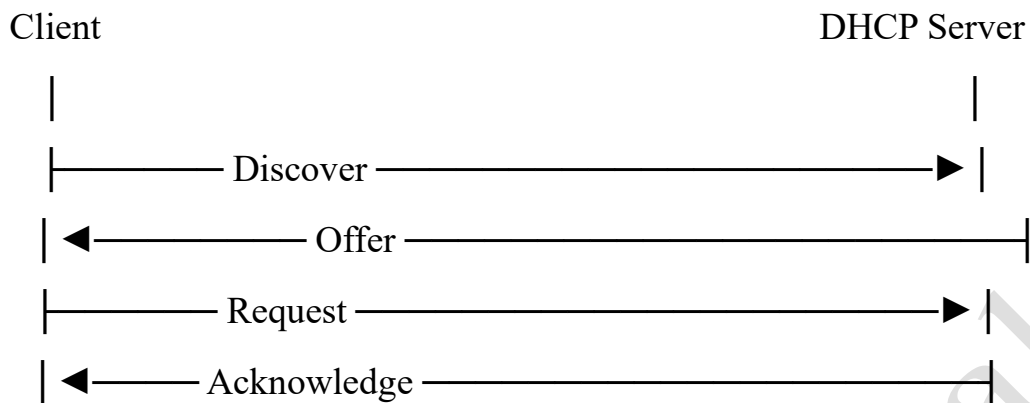
DHCP (Dynamic Host Configuration Protocol) is an **Application Layer** protocol that automatically assigns IP addresses and other network configuration parameters, such as the subnet mask, default gateway, and DNS server address, to devices connected to a network. It eliminates the need to manually configure network settings for each device.

Working of DHCP

The DHCP server assigns network configuration to a client through the following four-step process:

1. **Discover** – The client broadcasts a DHCP Discover message to locate a DHCP server.
2. **Offer** – The DHCP server responds with an available IP address and network configuration.
3. **Request** – The client requests the offered IP address.
4. **Acknowledge** – The DHCP server confirms the assignment, and the client begins using the assigned network configuration.

DHCP Communication (DORA Process)



Features of DHCP

- Automatically assigns IP addresses.
- Assigns subnet mask, default gateway, and DNS server information.
- Prevents duplicate IP addresses.
- Simplifies network administration.
- Supports dynamic allocation of IP addresses.

Advantages

- Reduces manual configuration effort.
- Minimizes configuration errors.
- Efficiently manages available IP addresses.
- Simplifies adding new devices to the network.
- Improves overall network management.

Applications

- Home networks.
- Educational institutions.
- Corporate networks.
- Wireless (Wi-Fi) networks.
- Enterprise network environments.

DNS (Domain Name System)

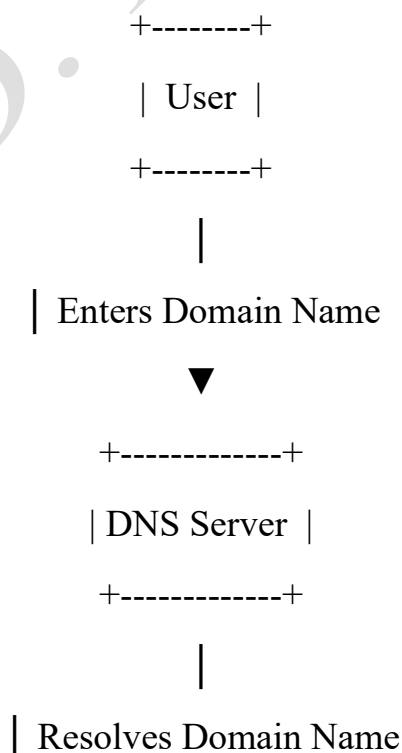
DNS (Domain Name System) is an **Application Layer** protocol that translates human-readable domain names into their corresponding IP addresses. It enables users to access websites using domain names instead of remembering numerical IP addresses.

For example, when a user enters www.google.com, the DNS server resolves the domain name into the corresponding IP address required to communicate with the destination server.

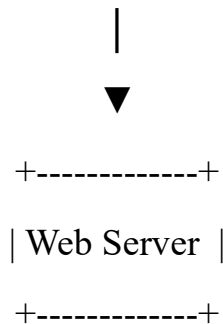
Working of DNS

1. The user enters a domain name in a web browser.
2. The request is sent to a DNS server.
3. The DNS server resolves the domain name into its corresponding IP address.
4. The IP address is returned to the user's device.
5. The browser establishes communication with the destination web server using the returned IP address.

DNS Communication



▼
Returns Corresponding IP Address



Features of DNS

- Converts domain names into IP addresses.
- Provides a hierarchical naming system.
- Supports distributed database architecture.
- Simplifies Internet access for users.
- Enables efficient domain name resolution.

Advantages

- Eliminates the need to remember numerical IP addresses.
- Makes Internet browsing simple and convenient.
- Supports fast domain name resolution.
- Reduces network management complexity.
- Improves user experience.

Applications

- Web browsing.
- E-mail services.
- Cloud services.
- Online applications.
- Internet-based communication.

Difference Between DHCP and DNS

Feature	DHCP	DNS
Full Form	Dynamic Host Configuration Protocol	Domain Name System
Layer	Application Layer	Application Layer
Primary Function	Automatically assigns IP addresses and network configuration	Resolves domain names into IP addresses
Purpose	Configures network devices automatically	Locates network resources using domain names
Information Managed	IP address, subnet mask, default gateway, and DNS server information	Domain names and their corresponding IP addresses
User Interaction	Automatic	Occurs when a domain name is accessed
Common Applications	Home, Office, Enterprise, and Wi-Fi Networks	Web Browsing, E-mail, Cloud Services, and Internet Applications

3.3 OSI/ISO Reference Model

Introduction

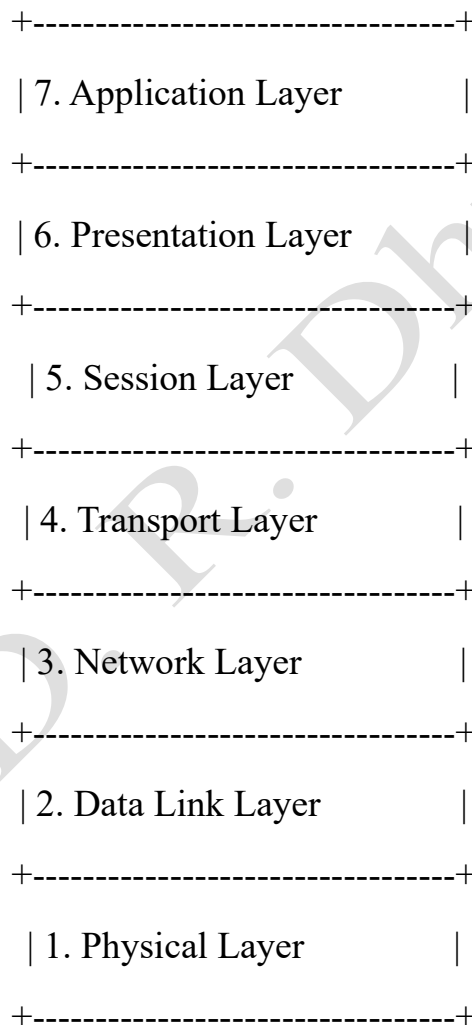
As computer networks grew in size and complexity, different manufacturers developed their own networking standards, making communication between devices difficult. To overcome this problem, the **International Organization for Standardization (ISO)** developed the **OSI (Open Systems Interconnection) Reference Model**.

The OSI Reference Model provides a standard framework for network communication by dividing the communication process into seven distinct layers. Each layer performs specific functions and communicates with the layers directly above and below it. This layered approach simplifies network design, implementation, troubleshooting, and maintenance.

OSI/ISO Reference Model

The **OSI (Open Systems Interconnection) Reference Model** is a conceptual framework developed by the **International Organization for Standardization (ISO)** that describes how data is transmitted between devices over a computer network. It divides the communication process into **seven layers**, where each layer performs specific networking functions and provides services to the layer above it.

Layers of the OSI Reference Model



1. Physical Layer

The **Physical Layer** is the lowest layer of the OSI model. It is responsible for transmitting raw bits over the physical communication medium. It defines the electrical, mechanical, and physical characteristics required for communication.

Functions

- Transmits raw bits between devices.
- Defines transmission media, cables, and connectors.
- Specifies electrical and signaling standards.
- Controls the transmission and reception of binary data.

Examples: Twisted Pair Cable, Coaxial Cable, Optical Fiber, Hub, Repeater

2. Data Link Layer

The **Data Link Layer** provides reliable node-to-node communication. It converts raw bits into frames and performs framing, physical addressing, flow control, and error detection to ensure reliable transmission.

Functions

- Performs framing of data.
- Uses MAC addresses for physical addressing.
- Performs error detection.
- Controls data flow.
- Provides reliable node-to-node communication.

Examples: Ethernet, Switch, MAC Address

3. Network Layer

The **Network Layer** is responsible for logical addressing and routing of packets between different networks. It determines the most efficient path for transmitting data from the source to the destination.

Functions

- Provides logical addressing.
- Performs routing.
- Selects the best transmission path.
- Transfers packets between different networks.

Examples: IP, Router

4. Transport Layer

The **Transport Layer** provides end-to-end communication between devices. It ensures reliable data delivery through segmentation, error recovery, and flow control.

Functions

- Provides end-to-end communication.
- Segments and reassembles data.
- Performs error detection and recovery.
- Controls data flow.
- Ensures reliable data delivery.

Examples: TCP, UDP

5. Session Layer

The **Session Layer** establishes, manages, synchronizes, and terminates communication sessions between applications. It also provides dialog control to ensure orderly communication between communicating devices.

Functions

- Establishes communication sessions.
- Provides dialog control.
- Synchronizes communication.
- Maintains active sessions.
- Terminates completed sessions.

Examples: NetBIOS, RPC

6. Presentation Layer

The **Presentation Layer** is responsible for data translation, encryption, and compression. It converts data into a common format that can be understood by different systems before transmission.

Functions

- Translates data formats.
- Performs encryption and decryption.
- Compresses and decompresses data.
- Converts data into a common standard format.

Examples: SSL/TLS, JPEG, ASCII

7. Application Layer

The **Application Layer** is the topmost layer of the OSI model. It provides network services directly to end-user applications and acts as the interface between users and the network.

Functions

- Provides network services to applications.
- Supports web browsing and e-mail services.
- Supports file transfer and remote access.
- Enables communication between users and network applications.

Examples: HTTP, FTP, SMTP, DNS

Advantages of the OSI Reference Model

- Provides a standard framework for network communication.
- Simplifies network design and implementation.
- Supports interoperability between different systems.
- Makes troubleshooting easier by separating functions into layers.
- Allows independent development and modification of each layer.

Disadvantages of the OSI Reference Model

- More complex than some practical networking models.
- Layered processing may increase communication overhead.
- Some functions overlap between layers.
- Primarily serves as a reference model and is not implemented exactly as defined in practical networks.

Protocol Data Units (PDU) of the OSI Model

OSI Layer	Protocol Data Unit (PDU)
Application Layer	Data
Presentation Layer	Data
Session Layer	Data
Transport Layer	Segment
Network Layer	Packet
Data Link Layer	Frame
Physical Layer	Bits

Summary of the OSI Layers

Layer No.	Layer Name	Primary Function	Examples
7	Application	Provides network services to users	HTTP, FTP, SMTP, DNS
6	Presentation	Data translation, encryption, compression	SSL/TLS, JPEG, ASCII
5	Session	Session establishment, dialog control, and synchronization	NetBIOS, RPC
4	Transport	End-to-end communication	TCP, UDP
3	Network	Logical addressing and routing	IP, Router
2	Data Link	Framing, physical addressing, flow control, and error detection	Ethernet, Switch
1	Physical	Transmission of raw bits through physical media	Twisted Pair Cable, Optical Fiber, Hub, Repeater

3.4 TCP/IP Reference Model

Introduction

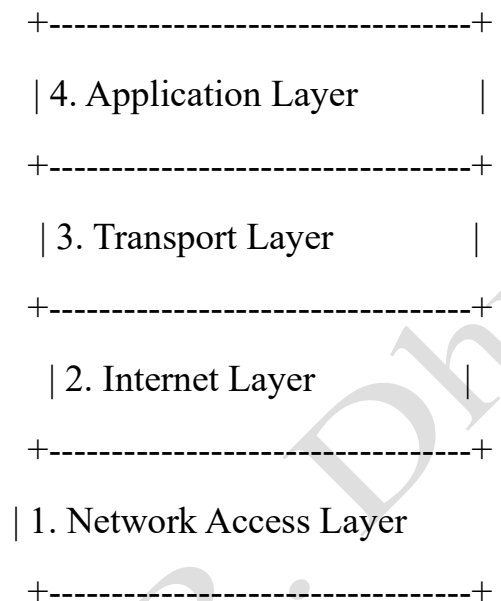
The **TCP/IP (Transmission Control Protocol/Internet Protocol) Reference Model** is the standard networking model used for communication over the Internet and modern computer networks. It was developed by the **United States Department of Defense (DoD)** to support reliable communication between interconnected networks.

Unlike the OSI Reference Model, which consists of seven layers, the TCP/IP Reference Model consists of **four layers**. Each layer performs a specific function and works together to ensure reliable communication between devices connected to a network.

TCP/IP Reference Model

The **TCP/IP Reference Model** is a conceptual networking model that describes how data is transmitted from one device to another over interconnected networks. It organizes communication into four layers, with each layer responsible for specific networking functions.

Layers of the TCP/IP Reference Model



1. Network Access Layer

The **Network Access Layer** is the lowest layer of the TCP/IP model. It is responsible for transmitting data over the physical network and defines how data is placed on the transmission medium.

Functions

- Transmits data over the physical network.
- Defines access to the transmission medium.
- Performs framing and physical addressing.
- Detects transmission errors.
- Supports communication between devices on the same network.

Examples: Ethernet, Wi-Fi, Switch, Network Interface Card (NIC)

2. Internet Layer

The **Internet Layer** is responsible for logical addressing and routing of packets between different networks. It determines the best path for transmitting data from the source to the destination.

Functions

- Provides logical addressing.
- Performs packet routing.
- Selects the best communication path.
- Transfers packets between interconnected networks.

Examples: IP, ICMP, Router

3. Transport Layer

The **Transport Layer** provides end-to-end communication between devices. It ensures reliable and efficient delivery of data using transport protocols.

Functions

- Provides end-to-end communication.
- Segments and reassembles data.
- Performs error detection and recovery.
- Controls data flow.
- Ensures reliable or connectionless data delivery.

Examples: TCP, UDP

4. Application Layer

The **Application Layer** is the topmost layer of the TCP/IP model. It provides network services directly to end-user applications and combines the functions of the Application, Presentation, and Session layers of the OSI model.

Functions

- Provides network services to applications.
- Supports web browsing, file transfer, and e-mail services.
- Enables remote login and domain name resolution.
- Provides services for communication between network applications.

Examples: HTTP, HTTPS, FTP, SMTP, DNS, DHCP

Advantages of the TCP/IP Reference Model

- Standard model for Internet communication.
- Supports communication between heterogeneous networks.
- Provides reliable and scalable communication.
- Flexible and widely implemented.
- Supports a large variety of networking applications.

Disadvantages of the TCP/IP Reference Model

- Does not clearly separate the Presentation and Session functions.
- Layer boundaries are less strictly defined than in the OSI model.
- More difficult to replace individual protocols.
- Primarily designed for the TCP/IP protocol suite.

Summary of the TCP/IP Layers

Layer No.	Layer Name	Primary Function	Examples
4	Application	Provides network services to end-user applications	HTTP, HTTPS, FTP, SMTP, DNS, DHCP
3	Transport	End-to-end communication, segmentation, and reliable data delivery	TCP, UDP
2	Internet	Logical addressing and routing of packets	IP, ICMP, Router
1	Network Access	Data transmission, framing, and physical addressing	Ethernet, Wi-Fi, Switch, NIC

Difference Between OSI and TCP/IP Reference Models

Feature	OSI Reference Model	TCP/IP Reference Model
Developed By	ISO	U.S. Department of Defense (DoD)
Number of Layers	7	4
Nature	Conceptual Reference Model	Practical Protocol Suite Model
Top Layer	Application Layer	Application Layer
Network Layer	Network Layer	Internet Layer
Lowest Layer	Physical Layer	Network Access Layer
Protocol Dependency	Protocol Independent	Based on the TCP/IP Protocol Suite
Primary Use	Learning, design, and standardization	Internet and modern computer networks

3.5 Switching – Circuit Switching, Packet Switching, and Message Switching

Introduction

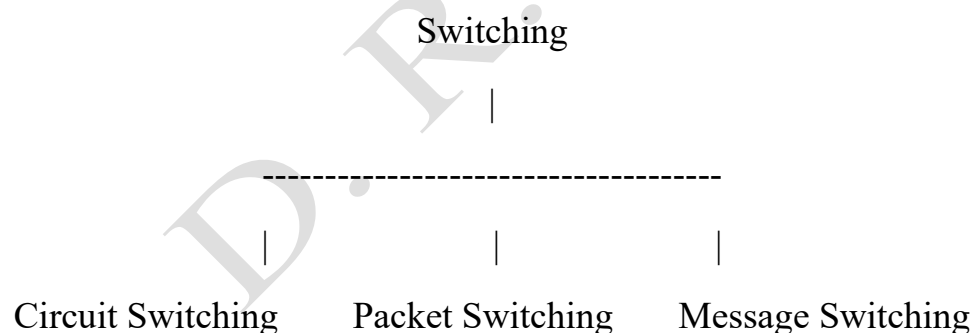
In a computer network, data must travel from the source device to the destination device through one or more intermediate nodes. The method used to transfer data from one network node to another is called **switching**. Switching enables efficient communication by selecting an appropriate path for data transmission and forwarding data through intermediate network devices.

Depending on the communication requirements and network design, different switching techniques are used. The three major switching techniques are **Circuit Switching, Packet Switching, and Message Switching**.

Switching

Switching is the process of transferring data from a source device to a destination device through a communication network by selecting an appropriate path. It enables multiple devices to communicate efficiently while making effective use of network resources.

Types of Switching



1. Circuit Switching

Circuit Switching is a switching technique in which a **dedicated communication path** is established between the sender and the receiver before data transmission begins. The communication path remains reserved exclusively for the entire communication session and is released only after the communication is completed.

Circuit Switching provides continuous and reliable communication because the entire bandwidth is reserved for the communicating devices.

Phases of Circuit Switching

1. Connection Establishment

A dedicated communication path is established between the sender and the receiver before data transmission begins.

2. Data Transfer

Once the connection is established, data is transmitted continuously through the reserved communication path.

3. Connection Termination

After communication is completed, the dedicated path is released and becomes available for other users.

Characteristics

- Establishes a dedicated communication path.
- Requires connection setup before data transmission.
- Uses the same communication path throughout the session.
- Reserves bandwidth for the entire communication.
- Suitable for continuous communication.

Advantages

- Reliable communication.
- Fixed bandwidth throughout the communication.
- Data is delivered in the correct order.
- Low transmission delay after connection establishment.

Disadvantages

- Inefficient utilization of bandwidth during idle periods.
- Connection establishment requires additional time.
- Failure of the dedicated path interrupts communication.
- Not suitable for bursty data traffic.

Applications

- Traditional Public Switched Telephone Network (PSTN)
- Integrated Services Digital Network (ISDN)
- Leased Line Communication

2. Packet Switching

Packet Switching is a switching technique in which a message is divided into **small units called packets**. Each packet contains both data and addressing information and is transmitted independently through the network. At the destination, all packets are reassembled to reconstruct the original message.

Packet Switching efficiently utilizes network resources because communication paths are shared among multiple users.

Types of Packet Switching

1. Datagram Packet Switching

In **Datagram Packet Switching**, each packet is treated independently. Packets may travel through different routes and may arrive at the destination in a different order.

Characteristics

- No dedicated communication path.
- Each packet is routed independently.
- Packets may follow different paths.
- Packets may arrive out of order.
- Used in Internet Protocol (IP) networks.

2. Virtual Circuit Packet Switching

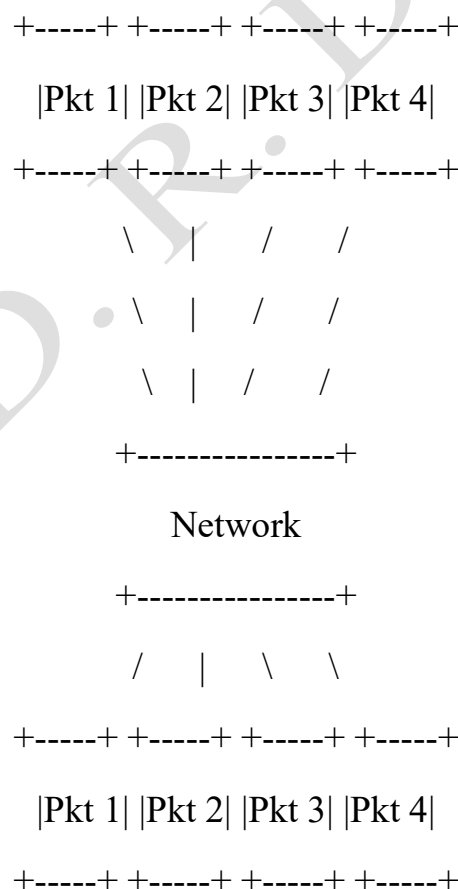
In **Virtual Circuit Packet Switching**, a logical communication path is established before data transmission begins. All packets follow the same path and usually arrive in the correct order.

Characteristics

- Logical path established before transmission.
- All packets follow the same route.
- Packets arrive in sequence.
- Provides more predictable communication.

Working of Packet Switching

Original Message



Packets are reassembled at the destination.

Characteristics

- Divides data into packets.
- Each packet contains addressing information.
- Packets may travel through different routes.
- Packets are reassembled at the destination.
- Efficient utilization of network bandwidth.

Advantages

- Efficient utilization of network resources.
- Supports communication among multiple users.
- Alternate paths improve reliability.
- Suitable for large computer networks.

Disadvantages

- Packets may arrive out of order.
- Packet delay may vary depending on network congestion.
- Reassembly is required at the destination.
- Network congestion may affect performance.

Applications

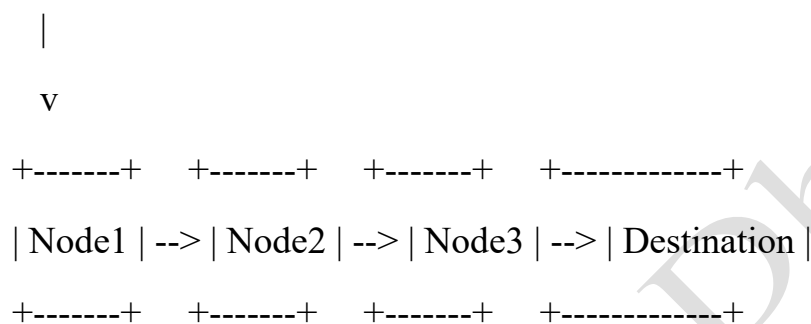
- Internet communication
- Web browsing
- Electronic mail
- File transfer
- Cloud computing services

3. Message Switching

Message Switching is a switching technique in which the **entire message** is transmitted as a single unit from one intermediate node to another. No dedicated communication path is established between the sender and the receiver. Each intermediate node temporarily stores the complete message before forwarding it to the next node. This method is known as the **Store-and-Forward Switching Technique**.

Working of Message Switching

Source



Each node stores the complete message before forwarding it to the next node.

Characteristics

- Entire message is transmitted as one unit.
- No dedicated communication path is required.
- Uses the Store-and-Forward Switching Technique.
- Intermediate nodes temporarily store messages.
- Suitable for non-real-time communication.

Advantages

- Efficient utilization of communication channels.
- No dedicated connection is required.
- Supports message prioritization.
- Reliable delivery through intermediate storage.

Disadvantages

- High transmission delay.
- Requires large storage at intermediate nodes.
- Unsuitable for real-time communication.
- Long messages occupy storage for a longer time.

Applications

- Telegraph systems
- Electronic Mail (Store-and-Forward)
- Message Handling Systems

Difference Between Circuit Switching, Packet Switching, and Message Switching

Feature	Circuit Switching	Packet Switching	Message Switching
Data Unit	Continuous data stream	Packets	Complete message
Dedicated Path	Yes	No	No
Connection Setup	Required	Not Required	Not Required
Transmission Method	Dedicated communication path	Independent packet transmission	Store-and-Forward Switching
Bandwidth Reservation	Reserved throughout communication	Shared dynamically	Shared dynamically
Storage at Intermediate Nodes	Not Required	Minimal	Required
Delivery Order	Sequential	May arrive out of order	Sequential

Delay	Low after connection setup	Variable	High
Resource Utilization	Low	High	Moderate
Suitable For	Real-time communication	Internet communication	Non-real-time communication
Examples	PSTN, ISDN	Internet	Telegraph, Electronic Mail

Important Points: TCP/IP and SMTP, DHCP and DNS with the DORA process, OSI/ISO Reference Model and the functions of its seven layers, TCP/IP Reference Model and the functions of its four layers, comparison between the OSI and TCP/IP models, Protocol Data Units (PDUs) used at each layer, Encapsulation and Decapsulation processes, Circuit Switching, Packet Switching (Datagram and Virtual Circuit), Message Switching (Store-and-Forward technique), and the comparison between Circuit Switching, Packet Switching, and Message Switching.

Module No. 4

Networking Devices and Advanced Networking

4.1 Network Devices – NIC Cards, Switches, Repeaters, Bridges, Gateways, and Routers

Network devices are specialized hardware components that enable communication between computers and other devices in a network. They play an essential role in transmitting data, directing network traffic, extending communication distance, connecting different networks, and ensuring efficient and reliable data exchange. Each network device performs a specific function based on the requirements of the network and the type of communication being established. The commonly used network devices are **Network Interface Card (NIC), Repeater, Bridge, Switch, Router, and Gateway**.

Network Interface Card (NIC)

A **Network Interface Card (NIC)** is a hardware component that enables a computer or any network-enabled device to connect to a network. It serves as an interface between the computer and the communication medium by converting digital data into electrical, optical, or radio signals for transmission and converting received signals back into digital data. Every NIC is assigned a unique **Media Access Control (MAC) Address**, which uniquely identifies the device within a Local Area Network (LAN). Depending on the type of connection, NICs are available as wired Ethernet cards or wireless Wi-Fi adapters. The NIC operates at the **Physical Layer** and the **Data Link Layer** of the OSI Reference Model.

The primary functions of a NIC are to establish network connectivity, provide a unique MAC address for device identification, convert digital data into network signals for transmission, receive incoming signals, and support communication over wired or wireless networks.

NICs are classified into two types:

- **Wired NIC (Ethernet Card):** Uses twisted pair or optical fibre cables for communication.
- **Wireless NIC (Wi-Fi Adapter):** Uses radio waves for wireless communication.

Advantages

- Enables communication between network devices.
- Provides unique device identification using a MAC address.
- Supports high-speed data communication.
- Easy to install and configure.

Disadvantages

- Failure of the NIC disconnects the device from the network.
- Wired NICs require physical cabling.
- Network performance depends on the capability of the NIC.

Repeater

A **Repeater** is a network device used to regenerate and retransmit weak or degraded signals over a communication medium. During data transmission, signals gradually lose their strength due to attenuation. A repeater receives these weakened signals, restores them to their original strength, and forwards them to the next segment of the network. It operates only at the **Physical Layer (Layer 1)** of the OSI Reference Model and does not examine or modify the transmitted data.

Computer ---- Weak Signal ---> Repeater ----- Strong Signal ---> Computer

The repeater extends the communication distance, regenerates signals, overcomes attenuation, and ensures that data reaches distant devices without significant signal loss.

Advantages

- Extends the communication range.
- Regenerates weak signals.
- Improves signal quality.
- Simple and economical networking device.

Disadvantages

- Does not filter network traffic.
- Cannot reduce network congestion.
- Operates only at the Physical Layer.

Bridge

A **Bridge** is a networking device that connects two or more similar Local Area Network (LAN) segments. It operates at the **Data Link Layer (Layer 2)** of the OSI Reference Model and forwards data frames using **MAC addresses**. Unlike a repeater, a bridge examines the destination MAC address before forwarding a frame, thereby filtering unnecessary traffic and improving network efficiency. Bridges are commonly used to divide large LANs into smaller collision domains, reducing network congestion and improving communication performance.

LAN Segment 1 ----- Bridge ----- LAN Segment 2

The bridge learns MAC addresses, filters data frames, forwards only the required traffic, and improves overall network performance.

Advantages

- Reduces unnecessary network traffic.
- Improves communication efficiency.
- Segments a LAN into smaller collision domains.
- Increases overall network performance.

Disadvantages

- Cannot connect networks using different protocols.
- Less suitable for large-scale networks.
- Slower than modern switches.

Switch

A **Switch** is an intelligent networking device that connects multiple computers and other devices within a Local Area Network (LAN). It operates primarily at the **Data Link Layer (Layer 2)** of the OSI Reference Model and forwards data frames using **MAC addresses**. A switch maintains a MAC address table to identify the devices connected to its ports and sends data only to the intended destination instead of broadcasting it to every device. This selective forwarding reduces collisions, improves bandwidth utilization, and significantly enhances network performance. Modern **Layer 3 switches** are also capable of performing limited routing functions.

A switch connects multiple devices, learns MAC addresses automatically, forwards frames to the correct destination, minimizes collisions, and supports efficient data communication.

Advantages

- Provides high-speed communication.
- Reduces network collisions.
- Improves bandwidth utilization.
- Supports full-duplex communication.
- Enhances network performance.

Disadvantages

- More expensive than a hub.
- Managed switches require configuration.
- Hardware failure affects connected devices.

Router

A **Router** is a networking device that connects two or more different networks and forwards data packets based on **Internet Protocol (IP) addresses**. It operates at the **Network Layer (Layer 3)** of the OSI Reference Model. A router examines the destination IP address of each packet and determines the best available path using **routing tables** and **routing algorithms**. Routers are widely used to connect local networks to the Internet and enable communication between geographically distributed networks.

LAN 1 ----- Router ----- Internet ----- Router ----- LAN 2

A router routes packets between networks, selects the most efficient communication path, connects local networks to the Internet, and controls data flow between interconnected networks.

Advantages

- Connects multiple networks.
- Supports Internet communication.
- Determines efficient routing paths.
- Improves network performance.
- Provides security features such as packet filtering.

Disadvantages

- More expensive than switches.
- Configuration is comparatively complex.
- Packet processing may introduce slight transmission delay.

Gateway

A **Gateway** is a networking device that connects two dissimilar networks using different communication protocols. It performs **protocol conversion**, enabling communication between systems that use different network architectures, data formats, and protocols. Unlike other networking devices, a gateway can operate at multiple layers of the OSI Reference Model depending on the type of translation or conversion being performed. It is commonly used to connect heterogeneous networks that would otherwise be unable to communicate with one another.

Network A (Protocol X)

|

Gateway

|

Network B (Protocol Y)

A gateway translates communication protocols, connects heterogeneous networks, enables interoperability between different systems, and allows seamless communication across diverse network environments.

Advantages

- Connects networks using different protocols.
- Performs protocol conversion.
- Enables interoperability between heterogeneous systems.
- Facilitates communication across different network architectures.

Disadvantages

- More complex than other networking devices.
- Requires expert configuration and maintenance.
- May become a communication bottleneck in large networks.

4.2 WiFi vs WiMAX

Wireless communication enables computers and other digital devices to exchange data without using physical cables. It has become an essential part of modern computer networks due to its flexibility, ease of installation, and mobility. Among the various wireless communication technologies, **WiFi** and **WiMAX** are widely used for providing wireless network connectivity. Although both technologies provide wireless Internet access, they differ in terms of coverage area, communication range, infrastructure, applications, and networking standards.

WiFi

WiFi is a wireless networking technology based on the **IEEE 802.11** standard. It enables computers, laptops, smartphones, tablets, printers, and other wireless devices to connect to a **Wireless Local Area Network (WLAN)** and access the Internet without using physical cables. WiFi communication is established through a **Wireless Access Point (AP)** or a **WiFi Router**, which allows multiple devices to communicate within a limited geographical area.

WiFi operates mainly in the **2.4 GHz, 5 GHz, and 6 GHz** frequency bands, depending on the IEEE 802.11 standard being used. It is commonly deployed in homes, offices, educational institutions, hospitals, airports, hotels, shopping malls, and public hotspots because of its high speed, low installation cost, and ease of deployment.

Advantages

- Provides high-speed wireless Internet access.
- Easy to install and configure.
- Eliminates the need for network cables.
- Supports multiple wireless devices simultaneously.
- Cost-effective for homes and offices.

Disadvantages

- Limited communication range.
- Signal strength decreases due to walls and obstacles.
- Performance may be affected by wireless interference.
- Requires proper security configuration to prevent unauthorized access.

WiMAX

WiMAX (Worldwide Interoperability for Microwave Access) is a broadband wireless communication technology based on the **IEEE 802.16** standard. It is designed to provide broadband wireless access over metropolitan and rural areas where wired communication infrastructure is unavailable, difficult to deploy, or expensive to install. WiMAX offers long-distance wireless communication and is commonly used by Internet Service Providers (ISPs) to deliver broadband Internet connectivity.

WiMAX generally operates in licensed and unlicensed microwave frequency bands and provides wireless communication over much larger distances than WiFi. It supports both fixed wireless access and mobile broadband communication, depending on the version of the standard.

Advantages

- Provides broadband connectivity over a large geographical area.
- Supports long-distance wireless communication.
- Suitable for both urban and rural areas.
- Supports a large number of users.
- Reduces the need for extensive wired infrastructure.

Disadvantages

- Higher installation and infrastructure cost.
- More complex to deploy and maintain.
- Requires licensed spectrum in many deployments.
- Less commonly deployed than WiFi in local networks.

Difference Between WiFi and WiMAX

Feature	WiFi	WiMAX
IEEE Standard	IEEE 802.11	IEEE 802.16
Network Type	Wireless Local Area Network (WLAN)	Wireless Metropolitan Area Network (WMAN)
Coverage Area	Up to approximately 100 metres	Up to approximately 50 km (depending on deployment conditions)
Communication Range	Short range	Long range
Frequency Band	2.4 GHz, 5 GHz, and 6 GHz	Licensed and unlicensed microwave frequency bands
Access Device	Wireless Access Point (AP) or WiFi Router	WiMAX Base Station
Infrastructure Cost	Low	High
Typical Speed	High over short distances	High over long distances
Primary Users	Homes, offices, schools, hotels, and public hotspots	Internet Service Providers, campuses, cities, and rural areas
Mobility	Local wireless access	Fixed wireless access and mobile broadband communication
Primary Application	Local wireless networking and Internet access	Metropolitan broadband wireless connectivity

4.3 Cloud Computing

Cloud Computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources such as servers, storage, databases, networking, software, and applications that can be rapidly provisioned and released with minimal management effort. Instead of purchasing, installing, and maintaining expensive hardware and software on local computers, users can access these resources over the Internet whenever required. This approach provides flexibility, scalability, reliability, and cost-effectiveness, making Cloud Computing one of the most important technologies in modern computing.

The term "**cloud**" refers to the Internet, which is commonly represented by a cloud symbol in network diagrams. In Cloud Computing, data and applications are stored and managed on remote servers maintained by cloud service providers rather than on a user's local computer. Users can access these resources anytime and from anywhere using devices such as desktops, laptops, tablets, and smartphones connected to the Internet.

Cloud service providers such as **Amazon Web Services (AWS)**, **Microsoft Azure**, and **Google Cloud Platform (GCP)** own and manage the cloud infrastructure. They provide computing resources to individuals, businesses, educational institutions, and government organizations on demand. Users pay only for the services they consume, eliminating the need for significant upfront investments in hardware and reducing the cost of infrastructure management.

Cloud Computing enables organizations to scale computing resources according to changing business requirements. Additional storage, processing power, or networking resources can be allocated within minutes without purchasing new hardware. This scalability, combined with high availability and global accessibility, has made Cloud Computing an essential technology for organizations of all sizes.

Characteristics of Cloud Computing

According to the **National Institute of Standards and Technology (NIST)**, Cloud Computing has five essential characteristics:

- **On-demand Self-Service:** Users can provision computing resources automatically without requiring human interaction from the service provider.
- **Broad Network Access:** Cloud services are accessible over the Internet using standard devices such as computers, laptops, tablets, and smartphones.
- **Resource Pooling:** Computing resources are shared among multiple users using a multi-tenant model while maintaining isolation and security.
- **Rapid Elasticity:** Resources can be quickly increased or decreased based on user requirements.
- **Measured Service:** Resource usage is monitored, measured, and billed according to actual consumption (Pay-as-you-use model).

Advantages of Cloud Computing

- Reduces the cost of purchasing and maintaining IT infrastructure.
- Provides scalable and flexible computing resources.
- Enables access to applications and data from anywhere with an Internet connection.
- Offers high availability and reliability.
- Simplifies data backup and disaster recovery.
- Supports collaboration among multiple users.
- Provides automatic software updates and maintenance.
- Allows organizations to deploy applications quickly.

Disadvantages of Cloud Computing

- Requires a stable and reliable Internet connection.
- Security and privacy concerns may arise if data is not properly protected.
- Users have limited control over the underlying cloud infrastructure.
- Service outages may temporarily affect access to applications and data.
- Vendor lock-in may make migration to another cloud provider difficult.
- Performance may vary depending on Internet bandwidth and network latency.

Applications of Cloud Computing

Cloud Computing is widely used in various sectors for delivering computing services efficiently. Some of its important applications include:

- Data storage and online backup.
- Web and mobile application hosting.
- Software development, testing, and deployment.
- Virtual Machines (VMs) and virtual servers.
- Online education and e-learning platforms.
- Business collaboration and productivity tools.
- Video streaming and multimedia services.
- Big Data analytics and Artificial Intelligence (AI).
- Internet of Things (IoT) applications.
- E-commerce and online business platforms.

Cloud Computing has transformed the way computing resources are delivered and utilized. By providing scalable, reliable, and cost-effective services over the Internet, it enables individuals and organizations to access powerful computing infrastructure without investing in expensive hardware. As digital transformation continues across industries, Cloud Computing remains a fundamental technology supporting innovation, collaboration, and efficient resource utilization.

4.4 Internet of Things (IoT)

The **Internet of Things (IoT)** is a network of physical objects, commonly referred to as "**things**," that are connected to the Internet and are capable of collecting, exchanging, and processing data with little or no human intervention. These objects are embedded with sensors, actuators, software, processors, and communication technologies that enable them to sense their surroundings, communicate with other devices, and perform specific tasks automatically.

IoT extends Internet connectivity beyond traditional computing devices such as computers and smartphones to a wide range of everyday objects, including home appliances, wearable devices, vehicles, industrial machines, medical equipment, and agricultural systems. By enabling these devices to communicate and exchange information, IoT helps automate processes, improve operational efficiency, optimize resource utilization, and support data-driven decision making.

An IoT system continuously collects data from connected devices, transmits the data through a communication network, processes the information using cloud or edge computing platforms, and provides meaningful insights or performs automated actions based on the analyzed data. As a result, IoT has become one of the most significant technologies driving digital transformation across various sectors.

Components of an IoT System

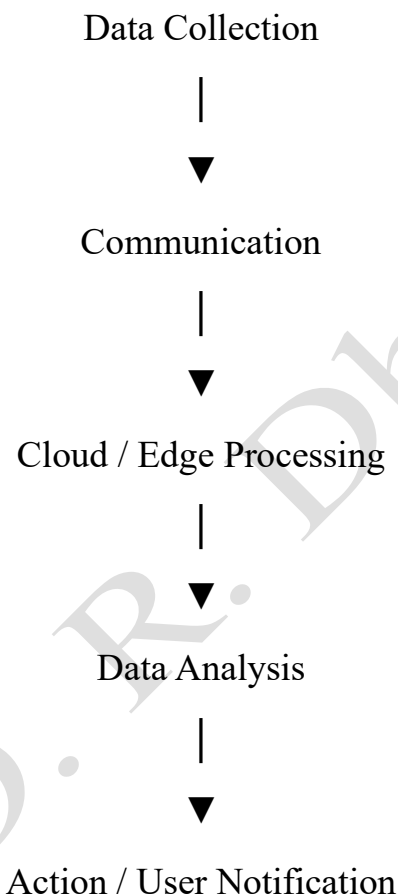
An IoT system consists of several interconnected components that work together to collect, transmit, process, and utilize data efficiently.

- **Devices (Things) and Sensors:** These devices collect real-time data such as temperature, humidity, pressure, motion, light intensity, and location. Some IoT devices also include actuators that perform actions based on received instructions.
- **Connectivity:** The collected data is transmitted through communication technologies such as Wi-Fi, Bluetooth, Zigbee, Ethernet, Cellular Networks, and LPWAN technologies.
- **Cloud or Edge Computing:** The received data is processed, analyzed, and stored to generate meaningful information and support intelligent decision making.

- **Application Interface:** Users interact with the IoT system through mobile applications, web portals, or dashboards to monitor devices, receive alerts, and control operations.

Working of IoT

The operation of an IoT system involves a sequence of activities that transform raw data into useful information and automated actions.



The working process can be summarized as follows:

1. Sensors collect real-time data from the surrounding environment.
2. The collected data is transmitted through a communication network.
3. Cloud or edge computing platforms process and analyze the received data.
4. Useful information is generated from the processed data.
5. The system performs automated actions or sends notifications and reports to users through applications or dashboards.

Advantages of IoT

- Enables automation of routine tasks.
- Supports real-time monitoring and control.
- Improves operational efficiency and productivity.
- Reduces human effort and operational costs.
- Facilitates remote monitoring and management.
- Enhances decision making through data analysis.
- Improves resource utilization and energy efficiency.
- Provides better customer experience through intelligent services.

Disadvantages of IoT

- Security and privacy risks due to Internet connectivity.
- Dependence on reliable network connectivity.
- Higher implementation and maintenance costs for large-scale deployments.
- Compatibility issues among devices from different manufacturers.
- Management becomes complex as the number of connected devices increases.
- Large volumes of data require efficient storage and processing.

Applications of IoT

IoT is widely used in numerous domains to improve automation, efficiency, and intelligent decision making.

- **Smart Homes:** Smart lighting, smart thermostats, smart locks, and home security systems.
- **Healthcare:** Remote patient monitoring, wearable health devices, and smart medical equipment.
- **Agriculture:** Smart irrigation systems, soil moisture monitoring, weather monitoring, and livestock management.

- **Industrial Automation:** Predictive maintenance, machine monitoring, production automation, and asset tracking.
- **Transportation:** Connected vehicles, traffic monitoring, fleet management, and smart parking systems.
- **Smart Cities:** Intelligent street lighting, waste management, water management, pollution monitoring, and public safety systems.
- **Retail:** Smart inventory management, automated billing, and customer behavior analysis.
- **Smart Energy Management:** Smart grids, smart electricity meters, and energy consumption monitoring.
- **Environmental Monitoring:** Air quality monitoring, weather forecasting, flood detection, and forest fire monitoring.

The Internet of Things has transformed the way physical devices interact with the digital world by enabling intelligent communication, automation, and real-time data exchange. With the rapid growth of connected devices and advancements in communication technologies, IoT is expected to play an increasingly important role in smart homes, healthcare, agriculture, manufacturing, transportation, and many other fields. Its ability to collect, analyze, and utilize real-time information makes IoT a fundamental technology in modern computer networks and digital transformation.

Thank You...